SCI-FLT-MATCHED v0.1

Optimal matched-template map filtering

Historical provisional title pending the explicit r0.6 owner disposition; `optimality_status=not_claimed`

Engineering Conformance Specification

CONDITIONAL-FREEZE PREFLIGHT – OWNER DISPOSITIONS PENDING. No numerical route is authorized. This document specifies evidence structure; it reports no implementation, conformity, response or covariance fidelity, observational validation, achieved performance, readiness, production, scientific freeze, or Unity result.

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Stage B r0.1 was derived exclusively from the owner-approved eight-object author packet; r0.2 applied the subsequent scientific-owner closure directive; r0.3 applies owner-authorized independent-review repairs without implementation input; r0.4 applies the second independent-review minor repairs and the owner's covariance-scope and diagnostic-only policy dispositions; r0.5 applied the owner's targeted type, lifecycle, covariance-role, and boundary closure; r0.6 applies only the owner's final stochastic-domain, lifecycle-order, route-multiplicity, role-label, and standalone-bundle micro-repairs while leaving title and AO authorization for explicit owner disposition. The admitted packet remains bound by SHA-256

`255c66da880fc7664a57635b28a98d874fc024490d04528f802635c0382a57c8`.

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Contents

Part I

Conformance frame

1 Purpose and claim discipline

This specification translates the shared scientific contract into auditable evidence obligations without naming or judging an implementation. A future conformity claim shall name one immutable realization, the exact frozen scientific-contract version, every scientific-owner disposition, engineering representation declaration and profile, the full validity domain, and the evidence bundle. A partial or aggregate pass cannot erase a failed requirement, contract consequence, option consequence, or named-use profile.

Conformity means that a realization implements the selected exact scientific operator and product semantics under one preregistered numerical-conformance profile. SCI-VAL separately evaluates immutable FLT-owned named-use profiles; observational/scientific validation is another evidence class. Performance measurement, readiness, and production are later layers. This draft supplies no result in any layer.

2 Required conformance-claim record

Before evaluation, a claim record shall freeze the following fields. Missing fields make the claim unavailable; evaluator faults make the attempt failed.

Field	Required content
Contract identity	`SCI-FLT-MATCHED v0.1`, shared-core digest, owner- disposed contract digest, and supersession status.
Realization identity	Complete realization tuple from Section 5, including parent class and state generation.
Dispositions	Exact scientific-owner choices, engineering preregistration/representation declarations, and successor-authorship triggers across `AO-001` through `AO-006`; no inferred defaults or representation-selected science.
Validity domain	Parent classes, WCS/frame, indexing, units, template and state strata, support/boundary classes, rank/null regimes, and exclusions.
Comparator	Exact $\mathcal{S}_{\text{parent fact}}$, $\mathcal{D}_{\text{model}}$, M, $\mathcal{D}_m$, m_{obs}, t_p, d_p, $\ell_p^{\star}$, c_p, $\mathcal{S}_{\text{apply}}$ and sparse amplitude reference identity, actual fixed-state producing map F_g, pre-draw frozen condition $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$, fixed covariance selector P_C, established linear operator identity when any, named covariance-role authority, exact subspace projectors, and $R^{[\delta,\sigma]}_{\text{realized}}$ response/state reconstruction identity.
Evidence identity	Immutable input, expected-result, observed-result, environment, tool/procedure, tolerance, uncertainty, and provenance records.
Verdicts	Requirement-, consequence-, option-, and profile-level outcomes using the complete lifecycle vocabulary, including distinct not requested, disabled, unavailable, learned candidate, resolved, applied, realized, complete publication candidate, publication decided, published, failed, not produced, and superseded semantics.
Nonclaims	Explicit statement that conformity does not establish validation, performance, readiness, production, or scientific authority.

3 Evidence rules

1. Every comparison shall bind exact quantity, unit, coordinate and index, support, state, population, aggregation norm, tolerance, and missing-value rule before observing the result.
2. Finite test sampling shall include a coverage or regularity argument that extends the observation to the declared validity domain. Unresolved output-affecting modes are failures or typed unavailability as specified by the

selected option.

3. Expected and observed results shall remain immutable and distinguish analytic identities, numerical comparisons, empirical sampling uncertainty, and profile-policy evaluation.
4. Any empirical covariance study shall preregister the sampling model, draw dependence or ergodic premise, finite-moment assumptions, covariance estimator and normalization, convergence mode/criterion, uncertainty or confidence construction, and coverage. These engineering choices do not alter the exact stochastic identities.
5. Operational response and covariance evidence shall predeclare fixed finite numerical domains. It shall record every failed/nonfinite evaluation or draw and shall not condition on success, censor, pairwise-delete, or change domains unless it evaluates a separately named population and role.
6. A test oracle shall not use the same undocumented operation as the realization under test. Shared scientific inputs are allowed; shared hidden state is not.
7. Unexpected errors, warnings that change scientific meaning, skipped required strata, and missing required data prevent a complete verdict.
8. Passing evidence establishes only its named claim for its named realization and domain. It cannot select an AO option or broaden authority.

Part II

Shared normative core

The following modules are imported verbatim by both contract views and are the sole normative source for definitions, equations, assumptions, requirements, contract consequences, option alternatives, edge cases, and failure semantics.

4 Normative notation and interpretation

4.1 Normative language

"Shall" is a contract requirement. "May" identifies an allowed form whose scientific preconditions still have to be met. "Exact" means equality in the declared mathematical realization. Ordinary finite-precision acceptance is an engineering numerical-conformance question and does not weaken an exact scientific identity. "Unknown" and "unavailable" never mean zero. An option in Section 10 is a decision candidate, not a selected route.

4.2 Collision-free objects, domains, and indexing

The paired readout-coordinate symbols x and r remain reserved to KID authority, and bare Q_p remains reserved to MAP/JINC authority. They are not used for this package's map-domain anchors or weighting.

A matched-template output anchor is p, a parent-map row or pixel is q, and a separately indexed template anchor is s. The base-v0.1 anchor set $\mathcal{D}_p$ contains exactly one anchor at each parent ordinary-MAP pixel center, with the same WCS, frame, grid, shape, indexing, ordering, and pixel-center convention as the parent apart from exact support restriction. The exact parent row-identity/fact domain is $\mathcal{S}_{\text{parent fact}}$. The exact coordinate domain of the authorized parent stochastic model is $\mathcal{D}_{\text{model}}$, and $M : \mathcal{D}_{\text{model}} \to \mathbb{R}$ is the parent random vector under one exact declared stochastic law. The observed numerical-payload domain is $\mathcal{D}_m = \{q \in \mathcal{S}_{\text{parent fact}} :$ an admitted finite real parent signal payload exists$\}$, and the immutable observed MAP payload $m_{\text{obs}} : \mathcal{D}_m \to \mathbb{R}$ exists only there. The expected parent response at row q per unit amplitude anchored at p is t_{pq}; its complete parent-coordinate vector is t_p^{full}. Before any local inversion or weight construction, the selected route declares the finite local operator domain $\mathcal{D}_{\text{loc}}(p)$ and its exact template extraction E_p, so $t_p = E_p t_p^{\text{full}}$. The scientific weighting W_p is the complete coordinate-basis bilinear operator on those restricted vectors. Only after W_p, d_p, and the row are constructed is the normalized local covector $\ell_p^\star = W_p t_p / d_p$ embedded into parent coordinates as $c_p = E_p^\top \ell_p^\star$. Application support is exactly $\mathcal{S}_{\text{apply}}(p) = \{q : c_{pq} \neq 0\}$. Exact zero is the canonical scientific representation; no numerical threshold defines support. The fixed-state row functional on any partially available numerical object is $L_{p,g} u = \sum_{q \in \mathcal{S}_{\text{apply}}(p)} c_{pq} u_q$. The dense shorthand $c_p^\top u$ is allowed only when u exists on the complete parent coordinate domain or an exact algebraic completion is proved irrelevant. The template matrix whose column s is t_s^{full} is T, and $R_{\text{fixed}}(p, s) = L_{p,g} t_s^{\text{full}}$.

All scientific input and output vectors are real. A transform representation may be complex only when it preserves real-field conjugate symmetry and uses the Hermitian adjoint. In the real parent-coordinate basis the adjoint is the ordinary transpose. No stationarity, isotropy, periodicity, flat-sky metric, or Euclidean physical metric follows from the vector notation.

Symbol	Normative meaning
p, q, s	Output anchor, parent row/pixel, and independently indexed template anchor.
$\mathcal{S}_{\text{parent fact}}$, $\mathcal{D}_{\text{model}}$, $\mathcal{D}_m$	Exact parent row-identity/fact domain, authorized stochastic-model coordinate domain, and the parent-fact subset carrying admitted finite real observed payload.
M, M_p	Parent random vector $M : \mathcal{D}_{\text{model}} \to \mathbb{R}$ under one exact declared stochastic law and its local restriction $M_p = E_p M$.
m_{obs}, $m_{\text{obs},q}$	Immutable observed ordinary-MAP payload and one sample, defined only on $\mathcal{D}_m$.
t_{pq}, t_p	Parent response per unit amplitude and its complete local-domain vector.
$\mathcal{D}_{\text{loc}}(p)$	Predeclared parent-coordinate domain used to construct the local template, weight, normalization, and operator before final application support is known.
E_p	Exact extraction/restriction from parent coordinates to $\mathcal{D}_{\text{loc}}(p)$.
$\mathcal{S}_{\text{apply}}(p)$	Parent payload coordinates whose final exact $L_{p,g}$ coefficients are nonzero.
$\ell_p^\star$, c_p	Normalized local covector $W_p t_p / d_p$ and its exact parent-coordinate embedding $E_p^\mathsf{T} \ell_p^\star$.
W_p	Frozen, self-adjoint, positive-semidefinite complete bilinear weight; all measure factors required by the scientific contraction are already in it.
n_p	Sparse weighted observed-parent contraction $\sum_{q \in \mathcal{S}_{\text{apply}}(p)} c_{pq} d_p m_{\text{obs},q}$.
d_p	Weighted template self-contraction $t_p^\mathsf{T} W_p t_p$; exactly real and positive for an admitted anchor.
$\widehat{a}_p$	Published fixed-template amplitude n_p/d_p.
$L_{p,g}$	Exact fixed-state scientific row functional producing $\widehat{a}_p$.
F_g	Actual fixed-state numerical producing map from an admitted observed payload, or the corresponding random input M when defining its stochastic law, to the realized amplitude field; it need not be exactly linear.
T	Full template-response operator over the declared template-anchor domain.
$R_{\text{fixed}}(p, s)$	Fixed-state response $L_{p,g} t_s^{\text{full}}$.
$R_{\text{FP}}(p, s)$	Full-procedure response when Learn–Resolve is rerun under an authorized perturbation definition; otherwise unavailable.
$R_{\text{realized}}^{[\delta,\sigma]}(m_{\text{obs}}; p, s)$	Operational fixed-state response of F_g at observed reference parent m_{obs} under the declared derivative or finite- difference specification. Here δ is the step (zero for an exact directional derivative) and σ is the derivative/sidedness scheme. A parent-independent matrix shorthand is allowed only after fixed-state linearity is established.
$C_{\text{parent}\mid\text{h}_{\text{pre}}}$	Authorized coordinate-basis covariance $\text{Cov}[M \mid h_{\text{pre}}]$ for one exact stochastic population conditional only on the pre-draw frozen condition h_{pre}.
C_p	Local-domain covariance $E_p C_{\text{parent}\mid\text{h}_{\text{pre}}} E_p^\mathsf{T}$.
$C_{U1,\text{reference}}$	Theorem/reference covariance $P_C L C_{\text{parent}\mid\text{h}_{\text{pre}}} L^\mathsf{T} P_C^\mathsf{T}$.
$C_{U1,\text{realized}}$	Operational conditional stochastic covariance $\text{Cov}[P_C F_g(M) \mid h_{\text{pre}}]$ on one fixed predeclared finite numerical codomain for the exact U1 population.
P_C	Fixed, predeclared selector from the produced field to the numerical codomain on which a named covariance role is defined; it is independent of the random draw and production outcome.
$\widetilde{d}_p$	Optional declared numerical normalization observable exposed by a numerical profile. It exists only when the producing route has a separable denominator with exact identity; it is not a required product of a general F_g.
θ	Frozen template/calibration state on which U1 conditional stochastic statements condition; variation over θ belongs to U2 rather than U1.
g_{resolved}	Immutable method/state generation after Resolve and before the random draw or Apply outcome.

Symbol	Normative meaning
$h_{\text{pre}} = (g_{\text{resolved}}, \theta)$	Pre-draw frozen condition containing only parent product role/identity/class, stochastic-law/population identity, template, local-domain/extraction, weight, support/missingness rules, subspace/rank/null, regularization, numerical profile, P_C, response-query domain, and failure semantics. Numerical m_{obs} values, a digest that fixes the draw, execution success, draw-dependent product identity, publication, censoring, pairwise deletion, and draw-dependent domains are not U1 conditioning variables.
$\mathcal{V}_{\text{signal}}$	Anchors satisfying every predicate required to publish the amplitude signal.
$\mathcal{V}_{\text{response}}(r)$, $\mathcal{V}_{\text{covariance}}(r)$	Signal-valid subdomains on which named response or covariance role r is valid.
$\mathcal{V}_{\text{NOI}}$, $\mathcal{V}_{\text{FRUIT handoff}}$	Signal-valid subdomain or bundle-level validity for the NOI child and FLT-to-FRUIT handoff roles.
`optimality_status`	One of `established_exact_local_GLS`, `not_claimed`, or `unavailable`; title, filename, and method ID are never evidence.
$\mathcal{Q}^{0.1}_{\text{FLT}}$	Finite FLT-owned state/response handoff query vocabulary for this contract version; queries outside it require a successor envelope.

4.3 Coordinate-basis measure and covariance convention

This package uses coordinate-basis column vectors and ordinary matrix multiplication. There is no additional measure-weighted bracket in the scientific estimator. Every fixed pixel area, quadrature weight, transform normalization, or metric factor required by a selected realization is part of W_p (and of the corresponding template/covariance coordinate convention).

For the parent random vector M with mean $\overline{M}$,

$$C_{\text{parent}} = \mathrm{E}[(M - \overline{M})(M - \overline{M})^{\mathsf{T}}], \qquad [C_{qq'}] = [M]^2.$$

It is a coordinate-basis covariance matrix, not an integral-kernel symbol. Under an invertible coordinate or quadrature reparameterization $M' = SM$ and $m'_{\text{obs}} = Sm_{\text{obs}}$ on their respective complete domains,

$$C' = SCS^{\mathsf{T}}, \qquad t' = St, \qquad W' = S^{-\mathsf{T}}WS^{-1},$$

so the corresponding complete-domain contractions and $t^{\mathsf{T}}Wt$ are invariant. A complex representation replaces transpose by Hermitian adjoint. Thus self-adjointness and positivity mean $W_p = W_p^{\dagger}$ and $u^{\dagger}W_p u \geq 0$ in the declared coordinate representation.

The units satisfy $[t] = [M]/[a] = [m_{\text{obs}}]/[a]$ and $[W] = [M]^{-2}$ for an exact covariance inverse; weaker weights may carry a common nonzero scale and different declared units because that scale cancels in n_p/d_p. In every case

$$[\widehat{a}] = [m_{\text{obs}}]/[t].$$

A point-source flux interpretation exists only if the template producer and CAL/BEAM lineage establish that exact amplitude convention. Otherwise the result is a declared shape amplitude.

5 Normative scientific definitions

5.1 Method, estimand, and fit boundary

`SCI-FLT-MATCHED` is a normalized matched-template map amplitude estimator. The word "optimal" applies only when the exact criterion and premises in Section 6.3 are established. It publishes a filtered amplitude map. At each fixed parent-pixel anchor it performs a fixed-template, fixed-anchor, one-parameter linear amplitude estimate. It does not select candidates, optimize position, fit morphology or background, deblend, infer source existence, or produce a catalog. It is not a posterior/Wiener sky reconstruction. Ordinary convolution does not acquire `SCI-FLT-MATCHED` identity or amplitude-estimator semantics merely through numerical resemblance. A selected realization may coincide numerically with a normalized correlation or convolution under proved special conditions; such coincidence does not weaken the fixed-template estimand or authorize generic convolution semantics.

Each realization has the immutable scientific identity

```
(method_id, estimand, parent_id, parent_class, stochastic_model_id, model_domain,
observed_payload_domain, anchor_lattice, WCS/frame, template_id,
requested/effective/resolved/applied/realized generation, weighting_object, exact_operator,
producing_map, established_linear_operator, numerical_profile, response_family, support_family,
units/calibration, covariance_scope, NOI_lifecycle, product_bundle, validity, failure_policy,
optimality_status, provenance).
```

A scientifically consequential change creates a new realization, generation, or separately named method. A representation-only change that exactly preserves the scientific object and supported query vocabulary does not change the estimand or scientific generation.

5.2 Parent classes and anchor lattice

Exactly one parent class is admitted per application: one immutable normalized ordinary-MAP observation bundle, or one immutable normalized ordinary-MAP coadd bundle with its exact contributing-observation membership. They are distinct applications. This contract asserts no filter/coadd commutation and owns no coaddition. JINC and other derived parents are not admitted. Numerical computability cannot repair missing MAP authority.

The base-v0.1 owner disposition places one output anchor at every exact parent ordinary-MAP pixel center. The output amplitude field preserves the parent's WCS, frame, grid, shape, indexing, order, phase origin, and pixel-center convention, except that exact support rules may make anchors unavailable. No interpolation is part of anchor creation. Template translation at an anchor, including even/odd sampling and ties, follows the exact deterministic template object. Subpixel, oversampled, and independent output lattices are successors.

5.3 Template-response scientific object

Each application binds one immutable template-response scientific object from exact owner-authorized source-class, morphology, beam, or physical-response authority. It binds the unit-amplitude convention, source-domain reference, parent response, anchor/WCS relation, sampling, phase, support, CAL/BEAM lineage, validity, generation, and supported query vocabulary.

The template shall not be learned, selected, centered, tuned, updated, or reparameterized from target MAP payload, a target-derived candidate or peak, the resulting amplitude field, target residuals, any NOI member, or a post-selection outcome. A predeclared template bank is admissible only when its member-selection rule is fixed independently of target outcomes. Selecting after inspecting target data defines a different inference method.

The same scientific object may be represented by exact materialization, an exact structured form, or exact lineage/query reconstruction. A representation-only change shall preserve every supported query and shall not alter the amplitude estimand. Template mismatch is not silently absorbed into normalization or parent covariance.

5.4 Five support roles

The following supports are distinct:

$\mathcal{S}_{\mathrm{apply}}(p)$.
Exact parent payload rows with nonzero coefficients in $L_{p,g}$. Exact-zero coefficients neither require nor dereference their payloads.

$\mathcal{S}_{\mathrm{template}}(p)$.
Exact rows and metadata required to define or query t_p.

$\mathcal{S}_{\mathrm{state}}(g)$.
Complete population influencing learned covariance, spectra, reliability, window, rank, null, regularization, or other resolved state.

$\mathcal{S}_{\mathrm{response}}(p, s)$.
Exact state, template, and operator support required by the declared response query.

$\mathcal{S}_{\text{store}}$.
Representation footprint. It has no scientific authority and may change losslessly.

The pre-inversion local operator domain $\mathcal{D}_{\text{loc}}(p)$ is declared before E_p, W_p, d_p, or $L_{p,g}$ is constructed. It is not inferred from the final exact-zero pattern. Coordinates in $\mathcal{D}_{\text{loc}}(p) \setminus \mathcal{S}_{\text{apply}}(p)$ may influence local covariance, weight, rank, normalization, or state and therefore remain in operator/state lineage even though Apply does not dereference their parent payload values. For an `SCI-FLT-MATCHED-AO-001-A` route, every $q \in \mathcal{D}_{\text{loc}}(p)$ shall lie in the authorized stochastic-model domain $\mathcal{D}_{\text{model}}$. Every $q \in \mathcal{S}_{\text{apply}}(p)$ shall lie in $\mathcal{D}_m$ and be admitted. A coordinate in $\mathcal{D}_{\text{loc}}(p) \setminus \mathcal{S}_{\text{apply}}(p)$ may participate in construction without any observed-payload dereference only when the exact template/model/covariance authority exists there. The shorthand $m_p = E_p m_{\text{obs}}$ is permitted only when a complete numerical local vector exists or when an explicit algebraic completion is proved irrelevant outside $\mathcal{S}_{\text{apply}}(p)$. An output anchor is admitted only when complete application support is present and valid. No partial-support estimator, edge renormalization, extension, imputation, background subtraction, learned taper, or signal-derived support belongs to this method. Numerical fill is allowed only after exact erosion proves zero application coefficient. Causal provenance retains $\mathcal{S}_{\text{state}}$ even when a fixed-state derivative touches only $\mathcal{S}_{\text{apply}}$.

5.5 Weighting and local covariance order

W_p is the selected frozen complete bilinear weight, exactly self-adjoint and positive semidefinite. A parent coefficient has only its selected-route role; its name, positivity, units, reciprocal, or diagonal placement does not make it covariance, precision, or inverse variance.

Local GLS first restricts the authorized parent covariance, conditional on h_{pre}, to the predeclared local domain: $C_p = E_p C_{\text{parent}|\text{h}_{\text{pre}}} E_p^{\mathsf{T}}$, then inverts it only on the exact declared local estimable subspace. This requires $\mathcal{D}_{\text{loc}}(p) \subseteq \mathcal{D}_{\text{model}}$, $M_p = E_p M$, and $C_p = \text{Cov}[M_p \mid h_{\text{pre}}]$. Missing model or covariance authority anywhere on $\mathcal{D}_{\text{loc}}(p)$ makes `SCI-FLT-MATCHED-AO-001-A`, its optimality status, and $v_{\text{GLS,reference}} = d_p^{-1}$ unavailable, even if a separately authorized sparse non-GLS amplitude remains computable. In general

$$(E_p C_{\text{parent}|\text{h}_{\text{pre}}} E_p^{\mathsf{T}})^{-1} \neq E_p C^{-1}_{\text{parent}|\text{h}_{\text{pre}}} E_p^{\mathsf{T}}.$$

A global-precision subblock is generally not the inverse of the locally marginal covariance. Under additional distributional assumptions it may equal a conditional precision, but using that block without the corresponding conditional mean and outside-data terms is not a complete conditional estimator. It is therefore a separately authored weight and method with a different influence domain, response, covariance, and edge rule.

5.6 Learn–Resolve–Apply state graph

State advances through three explicit operations:

Learn.
Derive candidate covariance, spectral/reliability information, window, rank, null, regularization, and diagnostics from one declared population, with source/target protection and state uncertainty recorded.

Resolve.
Select exactly one requested, owner-authorized `SCI-FLT-MATCHED-AO-001` method for this realization and bind its parameters, support, scientific identity, numerical-conformance policy, failure disposition, and immutable generation. Externally declared state enters here without observational learning.

Apply.
Use the resolved state unchanged. No target, candidate, source, or NOI-member update occurs during Apply.

The package may authorize multiple separately named `SCI-FLT-MATCHED-AO-001` methods, including both `SCI-FLT-MATCHED-AO-001-A` and `SCI-FLT-MATCHED-AO-001-C`, but one request and realization binds exactly one. There is no automatic selector, target-data-driven choice, route mixing, or fallback between them. Requested, effective, learned-candidate, resolved, applied, realized, complete-publication-candidate, publication-decided, and published identities are distinct even when their values happen to agree. Observation and coadd states are separate. A compatible NOI member receives the identical resolved transformation, support, normalization, and failure rules. Any update creates a successor; per-member relearning is a different future method.

For a successful requested product route, the exact order is

requested → effective → learned candidate
→ resolved → applied
→ realized → complete publication candidate → publication decided
→ (published or not produced),

with learned candidate omitted when Learn does not apply. not requested is an alternative initial disposition. disabled and unavailable are pre-application dispositions where applicable. failed may branch from Learn, Resolve, Apply, numerical realization, product closure, publication evaluation, or publication action.

5.7 NOI compatibility predicate

For a signal realization y and a candidate NOI member z, FLT evaluates a frozen admission predicate $K_{\mathrm{NOI}}(z, y)$ before Apply. The predicate uses only declared input facts: parent class and quantity; WCS, grid, shape, indexing, ordering, and pixel-center convention; units and calibration convention; template identity; resolved state generation; local-domain and missingness semantics; NOI population and membership authority; numerical profile; and failure policy. NOI owns population and membership meaning; FLT owns this transformation-compatibility decision. Compatibility never depends on whether a transformed value later succeeds. Missing required payload makes the affected member or anchor incompatible or unavailable; it never adapts support, changes normalization, or triggers member-specific Learn or Resolve.

5.8 Fixed-state, full-procedure, realized, and reference response

The fixed-state response is $R_{\mathrm{fixed}}(p, s) = L_{p,g_0} t_s^{\mathrm{full}}$. It may be asymmetric, anisotropic, nonstationary, position-dependent, or nonlocal. A learned procedure has a separately typed full-procedure response obtained only by an authorized exact rerun or finite-difference definition that reruns Learn and Resolve after perturbation. The state-change record is retained separately. Until such authority exists, R_{FP} is unavailable; no FLT, FRUIT, SRC, or other consumer may call R_{fixed} the full-procedure response.

The actual fixed-state numerical producing map is F_g. It acts on the observed payload m_{obs} for one numerical realization and on M when defining its stochastic law. It may include rounding, branches, exceptional-value handling, or typed attempt failure and is not presumed exactly linear. Its realized response is therefore the declared fixed-state directional derivative or predeclared finite difference of F_g without rerunning Learn or Resolve. The general response is $R^{[\delta,\sigma]}_{\mathrm{realized}}(m_{\mathrm{obs}}; p, s)$, where the response-query domain, reference parent, step δ, and derivative or sidedness scheme σ are fixed before evaluation. Every evaluation entering one derivative or finite difference shall return finite numerical values on the same predeclared output-query domain; otherwise that response query is unavailable. Conditioning on successful evaluations, censoring failed evaluations, or changing the domain defines a separately named response experiment. Only after evidence establishes $F_g(m_{\mathrm{obs}}) = \widetilde{L}_g m_{\mathrm{obs}}$ on the declared domain may the parent-independent matrix identity $R_{\mathrm{realized}} = \widetilde{L}_g T$ be used. The exact scientific comparator is $R_{\mathrm{reference}} = LT$. These are not interchangeable.

Every fixed-state expectation, reference variance, reference covariance, and realized covariance statement conditions only on the pre-draw frozen condition $h_{\mathrm{pre}} = (g_{\mathrm{resolved}}, \theta)$. It may contain the immutable parent product role, identity, and class; exact stochastic-law/population identity; template; $\mathcal{D}_{\mathrm{loc}}/E_p$; weight; support/missingness rules; subspace/rank/null; regularization; numerical profile; P_C; response-query domain; and failure semantics. It shall not contain numerical m_{obs} values, an observed-payload digest that fixes the draw, execution success, a draw-dependent realized-product identity, publication, censoring, pairwise deletion, or a draw-dependent domain. Those are provenance rather than U1 conditioning variables. Conditioning on observed target payload requires a separately named conditional population and covariance role. U2 describes variation over the frozen template/calibration state θ and is excluded from those U1 conditional statements. A named operational covariance role is defined only after a fixed, predeclared selector P_C establishes a finite numerical random-vector codomain. It exists only when $P_C F_g(M)$ is finite almost surely under the declared U1 law. Conditioning on successful production, censoring, pairwise deletion, or a draw-dependent domain is not the same covariance role and requires a separately named population and covariance contract.

5.9 Product roles and atomic lifecycle

The required atomic signal bundle is valid on $\mathcal{V}_{\text{signal}}$ and contains the filtered amplitude field plus its quantity/unit, parent and anchor-lattice identity, template convention, resolved and realized operator/state identity, application validity, actual operational-response reference, uncertainty availability, lifecycle, and complete provenance. Failure of any required member prevents a complete publication candidate.

Every signal bundle carries the identity, request status, validity domain, and lifecycle status of each named covariance role. Independently atomic qualified companions may carry an authorized available covariance scope, frozen-NOI uncertainty, response, or exact state/lineage. Every companion has its own scientific identity, support, population, unit, validity, lifecycle, and named-use policy. Under selected `SCI-FLT-MATCHED-AO-003-C`, typed operational-covariance unavailability does not make the signal bundle partial unless an exact named-use policy requires that companion. Optional failure does not change the signal estimand. No numerator, denominator, kernel, field-power estimate, response slice, inverse scale, standardized field, or intermediate is a public science product without a complete role and policy.

5.10 Lifecycle vocabulary

State	Meaning
not requested	No request exists. This is not disabled.
requested	A named method/product/use was requested; no admission is yet implied.
effective	Boundary resolution determined that the request applies to this parent and method identity.
learned candidate	Learn produced candidate state that has not been owner-authorized, frozen, or resolved for Apply.
disabled	An applicable requested route was explicitly disabled by its authorized state or policy.
unavailable	A required authority, input, option, support, or scientific premise is absent; no numerical value is asserted.
resolved	Method, state, parameters, support, policy, and failure disposition are immutable for the generation.
applied	The resolved operator was invoked on an admitted parent.
realized	The declared fixed-state producing map generated immutable values and outcome provenance; realization neither implies publication nor changes pre-draw conditioning.
complete publication candidate	Every required realized value, identity, status, validity fact, companion identity/status record, lifecycle member, and provenance member exists and is ready for publication-policy evaluation. This is not a finding of publication eligibility, validation, or readiness.
publication decided	The exact owner-policy/SCI-VAL decision artifact exists for one immutable complete candidate without changing its scientific content.
published	An authorized publication decision exposed the immutable product under its declared identity, role, validity, and lineage.
failed	An authorized attempt began but Learn, Resolve, Apply, numerical realization, product closure, publication evaluation, or publication action failed.
not produced	A complete disposition intentionally created no public product; it is not zero or unavailable.
superseded	An immutable successor exists under an authorized policy; the predecessor is retained and never rewritten.

5.11 SCI-VAL named-use evaluation

SCI-VAL evaluates an owner-approved, Registry-bound named-use profile. The role-semantics drafts returned with this package are unregistered and unevaluable; missing policy fields are never inferred. A registered profile retains four independent axes: request, applicability, eligibility, and realization. FLT owns the producer facts, dependency graph, named-use disposition, prescribed consumer action, and failure semantics. VAL registers and evaluates; it does not filter, publish, select an option, invent a missing fact, or perform observational/scientific validation. Each profile keeps separate verdicts for parent admission (PA), state admission (SA), signal publication (SP), conditional-covariance use (CU), NOI-companion use (NU), response-companion publication/use (RU), and FLT-to-FRUIT producer-envelope use (FH). The SCI-VAL publication-decision artifact is realized independently of the FLT signal product; it does not move FLT product realization earlier or mutate it.

Each signal bundle has exact MAP and template boundary identities. NOI and FH use additionally records exactly one boundary-request state: `not_requested`, `inapplicable`, `unavailable`, `compatible`, `requested_pending`, or `realized_child_exists`. A missing or unrequested optional child does not narrow $\mathcal{V}_{\mathrm{signal}}$ or block an unrelated role. A child never mutates its parent signal.

5.12 First-class FLT-to-FRUIT producer envelope

The FLT product contract preserves a first-class interface sufficient for a future FRUIT method to consume the filtered amplitude product without reconstructing undocumented FLT implementation state. The finite v0.1 FLT-owned handoff vocabulary $\mathcal{Q}_{\mathrm{FLT}}^{0.1}$ contains exactly: filtered quantity and per-anchor value/validity/lifecycle; parent and anchor identity/WCS; template identity and amplitude convention; local domain and application support; fixed-state scientific and available realized/reference response identities; method, resolved state, producing-map, and numerical- profile identities; named U1/U2 and companion availability; units/calibration; provenance; and the exact supported state/response query declarations. Selected `SCI-FLT-MATCHED-AO-004` representation is required for state reconstruction and selected `SCI-FLT-MATCHED-AO-005` representation for response reconstruction. A future FRUIT method requiring a query outside $\mathcal{Q}_{\mathrm{FLT}}^{0.1}$ requires a successor FLT envelope or generation; existing signal products remain unchanged. FLT defines no FRUIT model, recurrence, subtraction/add-back, learning, stopping, restart, selection, response, uncertainty, or interpretation.

The handoff role has its own $\mathcal{V}_{\mathrm{FRUIT\ handoff}}$, either as a subdomain of $\mathcal{V}_{\mathrm{signal}}$ or as a bundle-level predicate. A failed or unrequested handoff does not narrow signal validity.

5.13 Producer ownership and claim layers

MAP owns the immutable parent identity, support, response, covariance, beam, calibration, WCS, lifecycle, and provenance. The template producer owns the unit-amplitude response object. CAL/BEAM owns calibration and beam authority. FLT owns the exact transformation, resolved state, operational realized response/covariance consequences, products, lifecycle, finite handoff query vocabulary, and one-way producer envelope. NOI owns its realization population and empirical uncertainty meaning. FRUIT owns any future FRUIT science. No SRC boundary is created.

Scientific authority, implementation conformity, representation fidelity, observational/scientific validation, achieved performance, readiness, and production are separate claims. This draft establishes none beyond its own proposed scientific text.

6 Reference estimator, response, and uncertainty

6.1 Authoritative coordinate-basis operator

For every admitted anchor $p \in \mathcal{V}_{\text{signal}}$, first declare $\mathcal{D}_{\text{loc}}(p)$ and its extraction E_p, then construct

$$t_p = E_p t_p^{\text{full}}, \qquad d_p = t_p^{\mathsf{T}} W_p t_p, \tag{1}$$

$$\ell_p^{\star} = \frac{W_p t_p}{d_p}, \qquad c_p = E_p^{\mathsf{T}} \ell_p^{\star}, \tag{2}$$

$$\mathcal{S}_{\text{apply}}(p) = \{q : c_{pq} \neq 0\}, \qquad \widehat{a}_p(m_{\text{obs}}) = \sum_{q \in \mathcal{S}_{\text{apply}}(p)} c_{pq} m_{\text{obs},q}, \tag{3}$$

$$n_p = d_p \widehat{a}_p, \qquad L_{p,g} u = \sum_{q \in \mathcal{S}_{\text{apply}}(p)} c_{pq} u_q, \tag{4}$$

$$R_{\text{fixed}}(p,s) = L_{p,g} t_s^{\text{full}}. \tag{5}$$

$W_p = W_p^{\dagger}$ and $u^{\dagger} W_p u \geq 0$. The scientific normalization d_p is finite, exactly real, and strictly positive. Failure of one of these exact predicates is typed unavailability or attempt failure, never amplitude zero. Ordinary floating-point evidence for the exact identities belongs to the preregistered numerical-conformance profile. Every active $q \in \mathcal{S}_{\text{apply}}(p)$ shall be in $\mathcal{D}_m$ and admitted. Coordinates used only to construct t_p, W_p, d_p, or c_p need no observed-payload dereference when their final coefficient is exact zero. Exact zero is not a tolerance. The dense shorthand $L_{p,g} u = c_p^{\mathsf{T}} u$ or $m_p = E_p m_{\text{obs}}$ may be used only when the complete numerical vector exists or an explicit completion is algebraically irrelevant outside $\mathcal{S}_{\text{apply}}(p)$.

6.2 Matching-template and general-sky relations

For the matching single-template model

$$M = A t_s^{\text{full}} + N, \qquad \mathrm{E}[N \mid h_{\text{pre}}] = 0,$$

linearity gives

$$\mathrm{E}[\widehat{a}_p(M) \mid h_{\text{pre}}] = A R_{\text{fixed}}(p,s).$$

At $p = s$, exact normalization yields

$$R_{\text{fixed}}(p,p) = 1, \qquad \mathrm{E}[\widehat{a}_p(M) \mid h_{\text{pre}}] = A.$$

For a general linear sky and deterministic residual,

$$M = \sum_s A_s t_s^{\text{full}} + b + N,$$

the exact fixed-state expectation is

$$\mathrm{E}[\widehat{a}_p(M) \mid h_{\text{pre}}] = \sum_s R_{\text{fixed}}(p,s) A_s + L_{p,g} b. \tag{6}$$

Unit diagonal response does not make off-diagonal response vanish. Neighboring or nonlocal templates may contribute, and a mean, offset, gradient, or other unmodeled deterministic b may bias the one-parameter amplitude. These terms are response and nuisance leakage, not source attribution or deblending.

6.3 Local restriction and constrained GLS theorem

For an authorized parent stochastic model, `SCI-FLT-MATCHED-AO-001-A` requires $\mathcal{D}_{\text{loc}}(p) \subseteq \mathcal{D}_{\text{model}}$ and $M_p = E_p M$. Restriction of the authorized covariance to that predeclared local operator domain precedes inversion:

$$C_p = \mathrm{Cov}[M_p \mid h_{\text{pre}}] = E_p C_{\text{parent}|\text{h}_{\text{pre}}} E_p^{\mathsf{T}}. \tag{7}$$

Let the columns of U_p be an exact orthonormal basis, in this declared coordinate representation, for the authorized local row/covector subspace and $P_p = U_p U_p^\top$ its representation-specific projector. Define

$$C_{p,E} = U_p^\top C_p U_p, \qquad W_p = U_p C_{p,E}^{-1} U_p^\top, \qquad U_p^\top t_p \neq 0,$$

where $C_{p,E}$ is positive definite. This is not a claim that a restricted global inverse equals the inverse of the restricted covariance.

The competitor class contains exactly the linear estimates $\ell^\top M_p$ on the identical predeclared $\mathcal{D}_{\mathrm{loc}}(p)$, with the identical U_p and fixed state, $\ell = P_p \ell$, and unit response $\ell^\top t_p = 1$ to the identical projected template. The matched GLS row is

$$\ell_* = \frac{W_p t_p}{t_p^\top W_p t_p}.$$

For every competitor,

$$\mathrm{Var}(\ell^\top M_p \mid h_{\mathrm{pre}}) - \mathrm{Var}(\ell_*^\top M_p \mid h_{\mathrm{pre}}) = (\ell - \ell_*)^\top C_p (\ell - \ell_*) \geq 0,$$

and

$$v_{\mathrm{GLS,reference}}(p) = \mathrm{Var}(\widehat{a}_p \mid h_{\mathrm{pre}}) = d_p^{-1}.$$

There are no constraints beyond $\ell \in \mathrm{Range}(U_p)$ and $\ell^\top t_p = 1$ in the v0.1 theorem. This is the sole minimum-conditional-variance and d_p^{-1} variance route within that exact competitor class, not among every singular-covariance linear estimator. Covariance-null directions are outside this theorem unless separately authorized. Deliberately excluded or infinite-variance modes, weighting-null modes, and a projected template with no identifiable component remain distinct. A shared numerical null label or an unconstrained Moore–Penrose operation does not establish the theorem. Under an invertible local coordinate change S, the admissible row/covector subspace transforms as $S^{-\top} U_p$; orthonormality and $P_p = U_p U_p^\top$ are conveniences of the chosen representation, not invariant extra science.

Missing stochastic-model or covariance authority for any coordinate in $\mathcal{D}_{\mathrm{loc}}(p)$ makes this theorem, its optimality claim, and $v_{\mathrm{GLS,reference}} = d_p^{-1}$ unavailable. It does not by itself forbid a separately authorized non-GLS sparse amplitude whose exact active observed payload dependencies are present.

6.4 Fixed-state and full-procedure response

For frozen state g_0,

$$R_{\mathrm{fixed}}(p, s) = L_{p,g_0} t_s^{\mathrm{full}}.$$

If state was learned from the parent, a full-procedure response exists only under separately authorized rerun authority. With

$$g_\epsilon = \mathrm{Resolve}(\mathrm{Learn}(m_{\mathrm{obs}} + \epsilon t_s^{\mathrm{full}})),$$

it is the declared limit or predeclared finite difference

$$R_{\mathrm{FP}}(p, s) = \lim_{\epsilon \to 0} \frac{\widehat{a}_p(m_{\mathrm{obs}} + \epsilon t_s^{\mathrm{full}}; g_\epsilon) - \widehat{a}_p(m_{\mathrm{obs}}; g_0)}{\epsilon}, \tag{8}$$

or the exact finite-ϵ analogue with its step, sidedness, and domain fixed before evaluation. The state-change record is separate. Without this authority R_{FP} is unavailable.

6.5 Realized and reference response and covariance

Let L be the exact selected scientific operator and let the actual fixed-state numerical producing map be

$$y = F_g(m_{\mathrm{obs}}).$$

It is not presumed linear. For an admitted observed reference parent m_{obs}, its realized fixed-state response is the authorized directional derivative, written as the $\delta = 0$ derivative case,

$$R_{\mathrm{realized}}^{[0,\mathrm{D}]}(m_{\mathrm{obs}}; p, s) = \left[\mathrm{D}F_g[m_{\mathrm{obs}}] \, t_s^{\mathrm{full}} \right]_p, \tag{9}$$

or $R^{[\delta,\sigma]}_{\text{realized}}(m_{\text{obs}}; p, s)$ for a predeclared finite difference with step, sidedness, and output-query domain fixed before evaluation and with no Learn or Resolve rerun. Every required evaluation shall be finite on that same domain; otherwise the response query is unavailable. The U1 realized and reference covariance roles, conditional only on the pre-draw frozen condition $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$ and on the fixed covariance selector P_C, are

$$R_{\text{reference}} = LT, \qquad C_{U1,\text{reference}} = P_C L C_{\text{parent}|\mathrm{h_{pre}}} L^{\mathsf{T}} P_C^{\mathsf{T}}, \tag{10}$$

$$C_{U1,\text{realized}} = \mathrm{Cov}[P_C F_g(M) \mid h_{\text{pre}}]. \tag{11}$$

The realized covariance exists only when $P_C F_g(M)$ is a fixed-dimensional finite real random vector almost surely under the declared U1 law. Successful- production conditioning, censoring, pairwise deletion, or draw-dependent domains define separately named populations and covariance roles; none is a fallback for the expression above. Only after evidence establishes $F_g(M) = \widetilde{L}_g M$ under the declared law and the corresponding observed-domain identity on the declared domain may one use

$$R_{\text{realized}} = \widetilde{L}_g T.$$

$$C_{U1,\text{realized}} = P_C \widetilde{L}_g C_{\text{parent}|\mathrm{h_{pre}}} \widetilde{L}_g^{\mathsf{T}} P_C^{\mathsf{T}}. \tag{12}$$

An amplitude field generated by $F_g(m_{\text{obs}})$ shall carry the operational realized pair, never the reference pair as though it described actual values. Either U1 covariance exists only with matching parent-law authority. U2 calibration/template-scale uncertainty is a separate named role. A joint U1+U2 product exists only when its random variables, conditioning, cross-covariances, and pushforward are separately authorized. A diagonal representation does not assert independence, and omitted entries are unknown.

6.6 Reference-variance and operational-covariance compatibility

$v_{\text{GLS,reference}}(p) = d_p^{-1}$ is a theorem-supported marginal reference variance only for `SCI-FLT-MATCHED-AO-001-A` under every premise of Section 6.3. It is not the same role as an operational covariance companion and survives typed unavailability of that companion.

Weight / covariance choice	Reference marginal variance	Operational U1 covariance
`SCI-FLT-MATCHED-AO-001-A` + `SCI-FLT-MATCHED-AO-003-C`	May be available as $v_{\text{GLS,reference}}(p)$	Unavailable; no covariance-based standardization, draw, independence, or covariance-dependent use.
`SCI-FLT-MATCHED-AO-001-A` + complete/projected `SCI-FLT-MATCHED-AO-003`	May be available	Separately available on $\mathcal{V}_{\text{covariance}}(r)$ when all operational premises hold.
`SCI-FLT-MATCHED-AO-001-C` or another non-GLS route	d_p is normalization only; reference variance unavailable	Separately available or unavailable under its own role.
Missing or population-mismatched parent covariance	Unavailable	Unavailable.

6.7 Engineering approximation observables

The following are comparison observables, not owner-selected scientific thresholds:

$$e_d(p) = \frac{|\widetilde{d}_p - d_p|}{d_p}, \qquad e_a(p) = |R^{[\delta,\sigma]}_{\text{realized}}(m_{\text{obs}}; p, p) - 1|, \tag{13}$$

$$e_R = \|P_{\text{out}}(R^{[\delta,\sigma]}_{\text{realized}} - R_{\text{reference}}) P_{\text{anc}}^{\mathsf{T}}\|, \qquad e_F(m_{\text{obs}}) = \|P_{\text{out}}(F_g(m_{\text{obs}}) - L m_{\text{obs}})\|, \tag{14}$$

$$e_C = \|P_{\text{out}}(C_{U1,\text{realized}} - C_{U1,\text{reference}}) P_{\text{out}}^{\mathsf{T}}\|. \tag{15}$$

e_d is available only when the preregistered profile declares $\widetilde{d}_p$ as a separable realized numerical normalization with exact identity and provenance. A general F_g need not expose such a denominator; then e_d is typed unavailable and direct field/response observables carry the profile. $\widetilde{d}_p$ is never promoted to a public science product. The response projection includes its declared reference-parent domain. A linear-operator discrepancy norm may additionally be used only when the $\widetilde{L}_g$ premise above is established. One preregistered engineering numerical-conformance profile binds norms, scales, domains, projections, coverage, floating-point environment, exact subspace projectors, and acceptance rules before evaluation. No package-wide constant is scientific authority. If an authorized operator intentionally changes scientific response, support, null space, or estimand, it receives a separate method/realization identity and publishes that discrepancy.

6.8 Uncertainty budget and non-combinability

ID	Contribution	Contract treatment
U1	Parent stochastic covariance	Use $C_{U1,\text{realized}} = \text{Cov}[P_C F_g(M) \mid h_{\text{pre}}]$ for the exact authorized population, fixed codomain, and conditioning; matrix propagation requires established fixed-state linearity.
U2	Calibration/template scale	Preserve joint CAL/BEAM and parent-template authority; variation over θ and missing cross-covariance are outside U1 and are unavailable rather than zero.
U3	Learned-state uncertainty	Base products condition on frozen state. Marginalization over Learn–Resolve needs separate authority.
U4	Numerical or intentional operator discrepancy	Ordinary numerical error belongs to the preregistered engineering profile. Intentional scientific discrepancy changes method/realization identity.
U5	Covariance scope/representation	Declare complete, projected, or unavailable scope separately from exact representation.
U6	Frozen-NOI empirical uncertainty	A qualified companion with its own population and finite-sample meaning; it is not automatically parent covariance or physical noise.
U7	Template mismatch/unmodeled systematics	Outside the base matching- template covariance and separately reported.

No scalar total uncertainty combines these terms unless every term, cross-term, conditioning relation, unit, and population is authorized. Absence is unavailable, never zero.

7 Assumptions and validity domain

The unbiasedness, response, optimality, and covariance statements are valid only on the intersection of the following declared predicates. Every product shall identify which predicates were evaluated, by whose authority, and with what immutable evidence.

ID	Assumption or predicate
`ASM-001`	The parent is exactly one immutable normalized ordinary-MAP observation or coadd bundle with authoritative identity, generation, grouping, WCS/frame, support, units, calibration provenance, and coadd membership when applicable.
`ASM-002`	Parent, template, physical amplitude, filtered amplitude, the stochastic parent M, observed payload m_{obs}, and coordinate-basis covariance have declared real codomains. The parent and output anchor lattices have the owner-disposed common WCS/frame/grid/indexing relation. Coordinate or quadrature changes transform $M, m_{\text{obs}}, t, C, W$ according to Section 4.3; a complex representation preserves conjugacy and uses the Hermitian adjoint.

ID	Assumption or predicate
ASM-003	t_p is the one immutable template-response scientific object for the application, supplied by exact owner-authorized source-class, morphology, beam, or physical-response authority. It expresses expected parent response per unit declared amplitude, with fixed scale, anchor, phase, validity, lineage, and query vocabulary in an exact representation. It is not learned, selected, centered, tuned, updated, or reparameterized from target MAP payload, target-derived peaks/candidates, the output field, target residuals, NOI members, or post-selection outcomes.
ASM-004	$\mathcal{S}_{\text{parent fact}}$, $\mathcal{D}_{\text{model}}$, and $\mathcal{D}_m$ are distinct. $\mathcal{D}_{\text{loc}}(p)$ and E_p are declared before local inversion and operator construction; `SCI-FLT-MATCHED-AO-001-A` requires $\mathcal{D}_{\text{loc}}(p) \subseteq \mathcal{D}_{\text{model}}$ and complete model/covariance authority there; complete final $\mathcal{S}_{\text{apply}}(p)$ is present and valid, every active row lies in $\mathcal{D}_m$, and construction-only rows with exact-zero c_{pq} need no payload. No admitted output depends on fill, extension, imputation, partial support, edge renormalization, adaptive background, learned taper, or signal-derived support. Template, state, response, and storage support remain distinct.
ASM-005	W_p is the exact frozen, self-adjoint, positive-semidefinite complete coordinate-basis bilinear weight named by the selected `SCI-FLT-MATCHED-AO-001` alternative, including population, units, rank, null, regularization, window, support, and edge rules.
ASM-006	d_p is finite, exactly real, and strictly positive.
ASM-007	For amplitude unbiasedness, the random parent follows the declared model $M = At_s^{\text{full}} + N$, and N has zero conditional mean on the complete stochastic support given $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$ and does not induce target-dependent changes in the template or operator.
ASM-008	For GLS optimality, $C_{\text{parent}\mid \text{h}_{\text{pre}}} = \text{Cov}[M \mid h_{\text{pre}}]$ is the exact authorized covariance for the population on $\mathcal{D}_{\text{model}}$; $M_p = E_p M$ and C_p are restricted to the predeclared local domain before inversion; $C_{p,E}$ is positive definite on the exact authorized row/covector subspace; the projected template is identifiable; and competitors share that domain, subspace, fixed state, and unit-response constraint.
ASM-009	For covariance, the parent law, actual fixed-state producing map F_g, operational response, support, units, and pre-draw frozen condition $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$ refer to the same realization and generation. h_{pre} may bind the parent product role/identity/class and stochastic-law/population identity, but never numerical m_{obs} values, a digest fixing the draw, success, a draw-dependent product identity, publication, censoring, pairwise deletion, or a draw-dependent domain. Those are provenance rather than conditioning variables. A fixed predeclared P_C supplies one finite numerical covariance codomain, and $P_C F_g(M)$ is finite almost surely under the declared U1 law. Reference quantities and U1/U2 roles remain separate; variation over θ belongs to U2.
ASM-010	State follows the declared Learn–Resolve–Apply graph and is frozen before Apply. It is neither target-updated nor NOI-member relearned.
ASM-011	A candidate NOI member satisfies the frozen pre-Apply K_{NOI} predicate in Section 5.7 and receives the identical state, operator, support, normalization, numerical profile, and failure policy as the signal parent.
ASM-012	A realized numerical operator is assessed under one preregistered engineering numerical-conformance profile. An intentional scientific discrepancy has separate method/realization identity.
ASM-013	Each named covariance role has a separate scope and representation; response and state query vocabularies and representations satisfy selected `SCI-FLT-MATCHED-AO-003`–`SCI-FLT-MATCHED-AO-005` lifecycle rules and the finite $\mathcal{Q}_{\text{FLT}}^{0.1}$ handoff vocabulary.
ASM-014	Template and parent calibration lineages are compatible; any claimed point-source flux meaning or calibration covariance has explicit CAL/BEAM authority.
ASM-015	A method option, named-use policy, and lifecycle route are explicitly selected; there is no automatic selector, fallback, or silent substitution.

7.1 Validity table

Dimension	Admission evidence	Invalid or missing case	Required outcome
Parent identity	Exact bundle identity, class, generation, grouping, WCS/frame, support, unit, quantity, calibration lineage, and coadd membership	Missing, mutable, derived, or mixed parent	`unavailable` before application; no FLT repair or coaddition.
Application class	Observation or coadd declared independently	Assumed commutation or borrowed state between classes	`unavailable`; require a distinct realization.
Stochastic model	Exact $\mathcal{D}_{\mathrm{model}}$, random vector M, population, law, pre-draw conditioning, and covariance authority covering every required $\mathcal{D}_{\mathrm{loc}}(p)$ coordinate for `SCI-FLT-MATCHED-AO-001-A`	Missing coordinate, population/law mismatch, or payload-conditioned law without a separately named role	`SCI-FLT-MATCHED-AO-001-A`, optimality, and $v_{\mathrm{GLS,reference}} = d_p^{-1}$ `unavailable`; a separately authorized non-GLS sparse amplitude may be evaluated independently.
Output anchors	One anchor at each exact parent pixel center with identical parent WCS, frame, grid, shape, index, order, and pixel-center convention	Interpolated, subpixel, oversampled, independently sampled, or ambiguous phase/tie rule	`unavailable` under v0.1; require a separately named successor.
Template	Exact owner-authorized source-class/morphology/beam/ physical- response object per unit amplitude with scale, phase, anchor, sampling, lineage, validity, and query vocabulary	Learned, selected, centered, tuned, updated, or reparameterized from target payload, target peak/candidate, output, residual, NOI member, or post-selection outcome; mismatched chart, ambiguous scale, lossy representation, or missing lineage	`unavailable`; no inferred template.
Local domain and application support	Predeclared $\mathcal{D}_{\mathrm{loc}}(p)$ and E_p, exact c_p, complete nonzero-coefficient payload closure in $\mathcal{D}_m$, and exact-zero erosion proof without a threshold, separately typed from state, template, response, and storage support	Domain inferred from final zeros, or missing, invalid, filled, extended, imputed, or partially supported dependency	Fail Resolve for an invalid construction; mark affected Apply anchors `unavailable`; no partial estimate.
Weighting state	Selected `SCI-FLT-MATCHED-AO-001`, population, self-adjoint PSD operator, rank/null, regularization, coefficients, window and edge rules bound at Resolve	Unknown, unselected, incompatible, or changing W_p	Fail closed; no fallback.
Normalization	Finite, exactly real, strictly positive d_p	NaN, infinite, complex, zero, negative, or null-confounded	Location `unavailable` or attempted bundle `failed`; never write zero.
Numerical realization	One preregistered engineering conformance profile with the exact comparator, projectors, norms, scales, domain, and acceptance rules	Post-hoc profile, uncovered modes, support/subspace mismatch, or failed rule	`failed` for that profile; a scientific operator change requires distinct identity.
Response	Fixed, full-procedure, realized, and reference types kept distinct, with selected exact domain/query/representation contract	Undocumented kernel, assumed invariance, strengthened fixed-state meaning, or unreconstructable response	Response-dependent product or handoff `unavailable`.

Dimension	Admission evidence	Invalid or missing case	Required outcome
Units and calibration	$[\widehat{a}] = [m_{\mathrm{obs}}]/[t]$ and compatible joint lineage	Unjustified flux, beam, solid angle, DC, extended-source, or covariance meaning	Forbid the claim; required named use is unavailable.
Covariance	Matching authorized parent law, actual F_g, pre-draw frozen condition h_{pre}, fixed predeclared P_C, finite numerical random-vector codomain, named U1/U2 role, scientific scope, and exact selected representation	Missing population or conditioning, failed or nonfinite draws on the fixed domain, selection on successful draws, censoring, pairwise deletion, draw-dependent domains, unproved linear propagation, unnamed calibration terms, omitted correlations treated as zero, or d_p^{-1} used without constrained local GLS	Covariance role unavailable or companion failed; signal may remain complete publication candidate.
NOI parity	Pre-Apply K_{NOI} compatibility and identical frozen transformation	Per-member relearning, post-result admission, adapted support, or mixed generation	Member/anchor or NOI companion unavailable or failed; separate method required.
Lifecycle	Atomic membership, named-use policy, immutable state, and the ordered Applied–Realized–Complete–PublicationDecided–Published/NotProduced closure	Mixed status, publication candidate before realization, partial signal bundle, silent mutation, or overwritten predecessor	Bundle failed; retain predecessor and successor identities.
FLT to FRUIT	Finite $\mathcal{Q}_{\mathrm{FLT}}^{0.1}$, selected `SCI-FLT-MATCHED-AO-004` for state reconstruction where required, selected `SCI-FLT-MATCHED-AO-005` for response reconstruction, and documented lineage	Requires an unsupported future query, undocumented implementation state, or supplies FRUIT science	Handoff/query unavailable; require a successor envelope for new queries; signal status is unchanged.

The signal domain $\mathcal{V}_{\mathrm{signal}}$ is the intersection of the predicates required for the amplitude role. For each response or covariance role r, $\mathcal{V}_{\mathrm{response}}(r) \subseteq \mathcal{V}_{\mathrm{signal}}$ and $\mathcal{V}_{\mathrm{covariance}}(r) \subseteq \mathcal{V}_{\mathrm{signal}}$ contain only anchors where that companion's additional predicates hold. Likewise $\mathcal{V}_{\mathrm{NOI}} \subseteq \mathcal{V}_{\mathrm{signal}}$ and the subdomain or bundle-level $\mathcal{V}_{\mathrm{FRUIT\ handoff}}$ is no broader than signal validity. Bundle-level validity additionally requires the role's own atomic identity and lifecycle members. An optional companion that is not requested or is unavailable never narrows $\mathcal{V}_{\mathrm{signal}}$; consumer-specific validity may be narrower.

8 Normative requirements

Requirement	Normative statement
`SCI-FLT-MATCHED-REQ-001`	The method identity shall be `SCI-FLT-MATCHED`. Human title and optimality claim are separate: the owner shall select the title disposition, and every public boundary shall carry `optimality_status=established_exact_local_GLS`, `not_claimed`, or `unavailable`. Title, filename, and method ID are never evidence.
`SCI-FLT-MATCHED-REQ-002`	The sole signal estimand shall be the fixed-template amplitude at each admitted anchor; the public signal role is a filtered amplitude map.
`SCI-FLT-MATCHED-REQ-003`	Each application shall bind exactly one immutable normalized ordinary-MAP observation parent or one immutable normalized ordinary-MAP coadd parent. JINC and other derived parents are excluded.

Requirement	Normative statement
`SCI-FLT-MATCHED-REQ-004`	Observation and coadd applications shall remain distinct; no commutation, equivalence, state sharing, or FLT-owned coaddition is asserted.
`SCI-FLT-MATCHED-REQ-005`	Each application shall bind one immutable, exactly owner-authorized source-class/morphology/beam/physical-response template object per unit amplitude. Exact materialized, structured, and lineage/query representations are admissible only when scientifically equivalent.
`SCI-FLT-MATCHED-REQ-006`	The template shall not be learned, selected, centered, tuned, updated, or reparameterized from target MAP payload, a target-derived candidate/peak, the resulting amplitude field, target residuals, any NOI member, or post-selection outcomes. A predeclared bank is admissible only when selection is independent of target outcomes.
`SCI-FLT-MATCHED-REQ-007`	A realization shall bind collision-free $p/q/s/W_p$ notation, coordinate-basis algebra, $\mathcal{S}_{\text{parent fact}}$, $\mathcal{D}_m$, $\mathcal{D}_{\text{model}}$, stochastic parent M, observed payload m_{obs}, $\mathcal{D}_{\text{loc}}$, real codomains, WCS/frame, indexing, units, support families, boundaries, estimable subspace, rank/null, regularization, validity, and any complex-transform adjoint convention.
`SCI-FLT-MATCHED-REQ-008`	The authoritative estimator shall construct $t_p = E_p t_p^{\text{full}}$, $d_p = t_p^\top W_p t_p$, $\ell_p^\star = W_p t_p / d_p$, $c_p = E_p^\top \ell_p^\star$, and shall apply only $\widehat{a}_p(m_{\text{obs}}) = \sum_{q:c_{pq}\neq 0} c_{pq} m_{\text{obs},q}$, with every scientific measure factor inside W_p.
`SCI-FLT-MATCHED-REQ-009`	Every admitted W_p shall be exactly self-adjoint and positive semidefinite and every admitted d_p finite, exactly real, and strictly positive. Numerical tolerances do not alter these scientific predicates.
`SCI-FLT-MATCHED-REQ-010`	$\mathcal{D}_{\text{loc}}(p)$ and E_p shall be fixed before weight or operator construction; complete final $\mathcal{S}_{\text{apply}}(p)$ shall be required. For `SCI-FLT-MATCHED-AO-001-A`, $\mathcal{D}_{\text{loc}}(p) \subseteq \mathcal{D}_{\text{model}}$ and exact model/covariance authority shall cover the whole local domain. Every active row shall lie in $\mathcal{D}_m$. Construction-only rows with exact-zero final coefficient shall not require observed-payload dereference; no threshold may define exact zero. Partial support, renormalization, extension, imputation, background subtraction, learned taper, and signal-derived support are excluded.
`SCI-FLT-MATCHED-REQ-011`	Fixed-state, full-procedure, realized, and reference response shall remain separately typed. Required response queries shall be exactly available or reconstructable; a reduced kernel requires proved invariances and measure.
`SCI-FLT-MATCHED-REQ-012`	`established_exact_local_GLS`, the word "optimal," and $v_{\text{GLS,reference}}(p) = d_p^{-1}$ require the constrained local inverse-covariance GLS theorem: restriction before inversion, a positive- definite estimable covariance, identifiable projected template, identical local domain/subspace/state, the sole unit-response constraint, and the exact competitor class.
`SCI-FLT-MATCHED-REQ-013`	The package may authorize multiple separately named `SCI-FLT-MATCHED-AO-001` methods. Authorization shall be explicit by package, parent class, and named use; every request and realization shall bind exactly one method identity. Observation and coadd authorizations and generations remain separate. No option in this draft is a default.
`SCI-FLT-MATCHED-REQ-014`	A parent coefficient shall retain only its declared route-specific role; its name, sign, units, reciprocal, or diagonal form does not imply covariance, precision, inverse variance, or independence.
`SCI-FLT-MATCHED-REQ-015`	State shall follow Learn–Resolve–Apply. Not requested, requested, effective, disabled, unavailable, learned candidate, resolved, applied, realized, complete publication candidate, publication decided, published, failed, not produced, and superseded states remain distinct; Apply uses immutable state and performs no target/member update.
`SCI-FLT-MATCHED-REQ-016`	Observation and coadd state generations shall remain separate and shall not be substituted for one another.

Requirement	Normative statement
`SCI-FLT-MATCHED-REQ-017`	Every NOI member satisfying the frozen pre-Apply K_{NOI} predicate shall receive the identical resolved state, transformation, application support, normalization, numerical profile, and failure rules as the corresponding signal realization.
`SCI-FLT-MATCHED-REQ-018`	An NOI-informed update shall create an immutable successor; per-member relearning is a different, unavailable-in-v0.1 method.
`SCI-FLT-MATCHED-REQ-019`	One realization shall not mix or choose among authorized methods. The method shall have no automatic selector, target-data-driven route choice, or fallback. A requested unavailable route fails closed without silent substitution, including between `SCI-FLT-MATCHED-AO-001-A` and `SCI-FLT-MATCHED-AO-001-C`.
`SCI-FLT-MATCHED-REQ-020`	Every produced signal bundle shall carry the identity, request status, validity domain, and lifecycle status of each named covariance role. When `SCI-FLT-MATCHED-AO-003` selects an available scope and its required parent law and fixed-domain authority exist, the corresponding companion shall carry $C_{U1,\mathrm{realized}} = \mathrm{Cov}[P_C F_g(M) \mid h_{\mathrm{pre}}]$; $C_{U1,\mathrm{reference}} = P_C L C_{\mathrm{parent}\mid \mathrm{h_{pre}}} L^{\mathsf{T}} P_C^{\mathsf{T}}$ remains a separate comparator. Matrix propagation requires established fixed-state linearity.
`SCI-FLT-MATCHED-REQ-021`	$v_{\mathrm{GLS,reference}}(p) = d_p^{-1}$ shall be called marginal reference conditional variance only under every constrained local GLS premise; otherwise d_p is normalization. Operational-covariance unavailability does not erase this separately supported theorem result. Missing model or covariance authority on any required $\mathcal{D}_{\mathrm{loc}}(p)$ coordinate makes the theorem and reference variance unavailable.
`SCI-FLT-MATCHED-REQ-022`	Uncertainty availability and U1–U7 shall be truthful and role-separated. U1 conditional stochastic covariance, U2 calibration/template uncertainty, and any authorized joint product shall have distinct identities; all U1 statements condition only on $h_{\mathrm{pre}} = (g_{\mathrm{resolved}}, \theta)$ and variation over θ belongs to U2; numerical observed payload values or digests that fix the draw, execution success, draw-dependent product identity/domain, publication, censoring, and pairwise deletion shall not enter h_{pre}; missing covariance, cross-covariance, or entries are unavailable rather than zero or independent.
`SCI-FLT-MATCHED-REQ-023`	The public signal product shall be the role-complete atomic bundle defined in Section 5.9, valid on $\mathcal{V}_{\mathrm{signal}}$.
`SCI-FLT-MATCHED-REQ-024`	A complete publication candidate shall be formed only after realization and shall contain every required immutable realized value, identity, status, validity fact, companion identity/status record, lifecycle member, and provenance member. Failure of any required signal member shall prevent that candidate; a partial bundle shall not be published as complete.
`SCI-FLT-MATCHED-REQ-025`	Covariance, frozen-NOI uncertainty, response, and state/lineage companions shall be independently atomic and qualified by exact identity, population, unit, support, their own subdomain or bundle validity, lifecycle, request status, and named-use policy. Optional unavailability shall not narrow $\mathcal{V}_{\mathrm{signal}}$.
`SCI-FLT-MATCHED-REQ-026`	The lifecycle states in Section 5.10 shall remain distinct with the exact definitions in Section 5.10; in particular not requested is not disabled, realized is not published, realization precedes candidate closure, and a complete publication candidate is ready for policy evaluation rather than already publication-eligible, validated, or ready.
`SCI-FLT-MATCHED-REQ-027`	No numerator, denominator, kernel, field-power estimate, response slice, inverse scale, standardized field, or intermediate is public without a complete science-product role and policy.

Requirement	Normative statement
`SCI-FLT-MATCHED-REQ-028`	The product contract shall preserve the finite first-class $\mathcal{Q}^{0.1}_{\mathrm{FLT}}$ producer envelope in Section 5. A future FRUIT requirement outside that vocabulary requires a successor FLT envelope; it cannot retroactively broaden an existing product.
`SCI-FLT-MATCHED-REQ-029`	No FLT artifact or profile shall define FRUIT source modeling, subtraction/add-back, recurrence, learning, stopping, restart, selection, response, uncertainty, or interpretation.
`SCI-FLT-MATCHED-REQ-030`	The method performs fixed-template, fixed-anchor, one-parameter linear amplitude estimation. Candidate selection, position/morphology/ background fitting, deblending, existence inference, catalogs, significance, completeness, purity, and SRC ownership remain outside scope.
`SCI-FLT-MATCHED-REQ-031`	Posterior/Wiener sky reconstruction, generic ordinary-convolution semantics, data-thresholded destriping, adaptive edge/background methods, and the historical high-pass/delta case remain separate methods. Proved numerical coincidence with normalized correlation/convolution does not change the `SCI-FLT-MATCHED` estimand. `SCI-FLT-FIXED` remains protected.
`SCI-FLT-MATCHED-REQ-032`	Every affected route shall receive its exact scientific-owner, engineering preregistration/representation, or successor-authorship disposition before dependent use is available. Representation choice cannot select science, and no disposition can manufacture an unavailable parent, template, or authority.
`SCI-FLT-MATCHED-REQ-033`	The scientific method is the exact selected n_p/d_p operator. Ordinary finite precision is evaluated under one preregistered engineering numerical-conformance profile. A scientifically different operator receives a separate method/realization identity and retains its discrepancy.
`SCI-FLT-MATCHED-REQ-034`	Scientific authority, implementation conformity, representation fidelity, observational validation, achieved performance, readiness, and production shall be separate claims with separate evidence and owners.
`SCI-FLT-MATCHED-REQ-035`	SCI-VAL shall evaluate an owner-approved named-use profile on separate request, applicability, eligibility, and realization axes. It shall not perform filtering, publication, or observational/scientific validation.
`SCI-FLT-MATCHED-REQ-036`	Parent and template calibration dependence shall remain joint. Point-source flux, matched-filter beam/solid angle, extended-source fidelity, and calibration covariance require exact separate authority.
`SCI-FLT-MATCHED-REQ-037`	A coadd application shall retain exact contributing-observation membership and parent provenance without asserting filter/coadd commutation.
`SCI-FLT-MATCHED-REQ-038`	Every signal product shall bind exact MAP and template boundary identities. NOI and FLT-to-FRUIT records shall carry exact identity and one named-use request state—`not_requested`, `inapplicable`, `unavailable`, `compatible`, `requested_pending`, or `realized_child_exists`—including K_{NOI} and $\mathcal{Q}^{0.1}_{\mathrm{FLT}}$ where applicable; the full realization tuple; immutable provenance; supported query/reconstruction contract; and a route-status record.
`SCI-FLT-MATCHED-REQ-039`	This Stage B draft and every unselected option shall make no implementation, conformity, covariance/response fidelity, observational validation, performance, readiness, production, scientific-freeze, or Unity claim.
`SCI-FLT-MATCHED-REQ-040`	Base v0.1 shall evaluate one anchor at each exact parent pixel center and preserve parent WCS/frame/grid/shape/index/order/pixel-center conventions. No interpolation is admitted; other lattices are successors.
`SCI-FLT-MATCHED-REQ-041`	For `SCI-FLT-MATCHED-AO-001-A`, $\mathcal{D}_{\mathrm{loc}}(p) \subseteq \mathcal{D}_{\mathrm{model}}$, $M_p = E_p M$, and local covariance shall be $C_p = \mathrm{Cov}[M_p \mid h_{\mathrm{pre}}] = E_p C_{\mathrm{parent}\mid \mathrm{h_{pre}}} E_p^{\mathsf{T}}$ on the predeclared $\mathcal{D}_{\mathrm{loc}}(p)$, and local GLS shall invert only its exact authorized subspace. A global-precision subblock is neither that marginal inverse nor a complete conditional estimator without its outside-data and conditional-mean terms.

Requirement	Normative statement
`SCI-FLT-MATCHED-REQ-042`	$\mathcal{S}_{\text{parent fact}}$, $\mathcal{D}_{\text{model}}$, $\mathcal{D}_m$, M, m_{obs}, $\mathcal{D}_{\text{loc}}$, $\mathcal{S}_{\text{apply}}$, $\mathcal{S}_{\text{template}}$, $\mathcal{S}_{\text{state}}$, $\mathcal{S}_{\text{response}}$, and $\mathcal{S}_{\text{store}}$ shall be separately recorded. Final exact-zero application coefficients do not dereference parent payload but retain causal operator/state lineage when they influenced construction.
`SCI-FLT-MATCHED-REQ-043`	The general-sky expectation shall retain all off-diagonal template response and $L_{p,g}b$ nuisance leakage. Unit diagonal response shall not imply source attribution, isolation, or absence of bias.
`SCI-FLT-MATCHED-REQ-044`	When state is parent-learned, full-procedure response requires an authorized Learn–Resolve rerun or predeclared finite difference and a separate state-change record; otherwise it is unavailable.
`SCI-FLT-MATCHED-REQ-045`	Any amplitude field generated by $F_g(m_{\text{obs}})$ shall carry its operational fixed-state realized response definition $R^{[\delta,\sigma]}_{\text{realized}}(m_{\text{obs}}; p, s)$ and any authorized $C_{U1,\text{realized}}$, not reference quantities mislabeled as actual. Response finite differences and covariance shall use fixed predeclared finite numerical domains with no success conditioning, censoring, pairwise deletion, or draw/evaluation-dependent domain. $\widetilde{L}_g T$ and matrix covariance propagation are admissible only when $F_g(M) = \widetilde{L}_g M$ and the corresponding observed-domain identity are established on the declared domains.
`SCI-FLT-MATCHED-REQ-046`	For each named covariance role, scientific scope shall be selected independently as complete within that role, named projected, or unavailable; exact representation shall be selected independently and shall not change the role or covariance identity. `SCI-FLT-MATCHED-AO-003-C` shall carry typed role identity and unavailability and shall not make the signal bundle partial unless an exact named-use policy requires that companion. It denies operational covariance and covariance-derived uses, not a separately theorem-supported $v_{\text{GLS,reference}}$ under `SCI-FLT-MATCHED-AO-001-A`.
`SCI-FLT-MATCHED-REQ-047`	Exact immutable state, its finite required query vocabulary, and its own role validity shall be reproducible across full, compact, or lineage representation without changing the estimator or signal generation. Representation is an engineering selection after the scientific vocabulary is fixed.
`SCI-FLT-MATCHED-REQ-048`	Response domain, finite query vocabulary, $\mathcal{V}_{\text{response}}(r)$, and consumer scope shall be defined before engineering chooses materialized, structured, or lineage representation. $\mathcal{Q}^{0.1}_{\text{FLT}}$ is the complete current handoff vocabulary; FLT shall not guess future FRUIT requirements.
`SCI-FLT-MATCHED-REQ-049`	PA, SA, SP, CU, NU, RU, and FH verdict meanings, separately addressable request/status records, role validity, and dependency graph are normative. An optional unrequested role shall not block unrelated SP. Profile count/layer/vector layout is lossless representation only; a scalar verdict shall not drive scientific action.
`SCI-FLT-MATCHED-REQ-050`	Covariance zero-variance modes, excluded or infinite-variance modes, weighting-null modes, and template-unidentifiable modes shall remain distinct exact subspace classifications.

9 Falsifiable contract consequences

Stable `SCI-FLT-MATCHED-PRED-` identities cover three explicitly typed classes: analytic scientific consequences, stochastic identities, and lifecycle/conformance invariants. Each is conditional on the cited assumptions and is falsified by a conforming counterexample in its declared domain. Empirical estimators and finite-sample acceptance belong to a preregistered engineering protocol. Failure authorizes no fallback.

The analytic class is `SCI-FLT-MATCHED-PRED-001`, `SCI-FLT-MATCHED-PRED-002`, `SCI-FLT-MATCHED-PRED-014`, and `SCI-FLT-MATCHED-PRED-019`–`SCI-FLT-MATCHED-PRED-022`. The stochastic class is `SCI-FLT-MATCHED-PRED-003`, `SCI-FLT-MATCHED-PRED-008`, and `SCI-FLT-MATCHED-PRED-024`. Every remaining `SCI-FLT-MATCHED-PRED-` identity, including appended `SCI-FLT-MATCHED-PRED-025`, is a lifecycle/conformance invariant. These type assignments are normative and do not change the stable IDs.

Consequence ID	Falsifiable statement	Assumptions
SCI-FLT-MATCHED-PRED-001	An exact unit-amplitude matching template at admitted p gives $R_{\text{fixed}}(p, p) = 1$.	003–006
SCI-FLT-MATCHED-PRED-002	At fixed state, a noiseless matching template scaled by A gives exactly $\widehat{a}_p = A$.	003–006
SCI-FLT-MATCHED-PRED-003	With zero conditional-mean noise and a matching template, $\mathrm{E}[\widehat{a}_p(M) \mid h_{\text{pre}}] = A$.	003–007
SCI-FLT-MATCHED-PRED-004	An anchor missing any nonzero application-support dependency is never published as a numerical amplitude.	004,006
SCI-FLT-MATCHED-PRED-005	Nonfinite, nonreal, zero, or negative d_p never produces zero amplitude as a substitute.	005–006
SCI-FLT-MATCHED-PRED-006	Observation and coadd applications retain distinct parent and state identities.	001,010
SCI-FLT-MATCHED-PRED-007	Every NOI member admitted by the frozen K_{NOI} predicate has the identical fixed L, support, normalization, profile, and failure decisions as its signal realization.	010–011
SCI-FLT-MATCHED-PRED-008	For the exact scientific fixed-state linear map and authorized parent law, $\mathrm{Cov}[P_C LM \mid h_{\text{pre}}] = P_C L C_{\text{parent}\mid \text{h}_{\text{pre}}} L^{\mathsf{T}} P_C^{\mathsf{T}}$. An NOI ensemble targets this identity only when separate authority establishes the same population, conditioning, fixed codomain, and covariance role.	001, 002, 004, 009, 013
SCI-FLT-MATCHED-PRED-009	Without exact constrained local GLS premises, removing a d_p^{-1} reference-variance label changes no amplitude value and removes the unsupported uncertainty claim. With those premises, typed operational- covariance unavailability does not remove the separately supported reference variance.	005,008–009
SCI-FLT-MATCHED-PRED-010	Acceptance under a preregistered engineering numerical profile implies only that every predeclared profile rule was satisfied; it is a conformance invariant, not a scientific error budget or performance claim.	012–013
SCI-FLT-MATCHED-PRED-011	A consequential realization-field change creates a new identity and never mutates an existing product.	010, 013, 015
SCI-FLT-MATCHED-PRED-012	Failure of one required signal member, or attempted candidate closure before realized immutable values exist, prevents a complete publication candidate.	015
SCI-FLT-MATCHED-PRED-013	Failure of an optional companion leaves a complete signal bundle unchanged unless its named-use policy requires that companion.	009, 013, 015
SCI-FLT-MATCHED-PRED-014	For a nonzero real pure amplitude-coordinate reparameterization $t \mapsto ct$ that holds $\mathcal{D}_{\text{loc}}$, E_p, W_p, g, support, and all non-template state fixed, $\widehat{a} \mapsto \widehat{a}/c$ while $\widehat{a}t$ is invariant. Any re-Learn or re-Resolve operation is a new realization outside this consequence.	003–006,014
SCI-FLT-MATCHED-PRED-015	A one-kernel response representation fails where its required invariances do not hold.	002,013
SCI-FLT-MATCHED-PRED-016	Requesting an unavailable route never silently selects another route.	015
SCI-FLT-MATCHED-PRED-017	A signal bundle plus its declared response/state interface answers every query in $\mathcal{Q}_{\text{FLT}}^{0.1}$ without undocumented state and returns typed unavailability outside it.	013,015
SCI-FLT-MATCHED-PRED-018	No conforming output alone is a detection, catalog entry, posterior sky, FRUIT result, performance claim, readiness decision, or production decision.	001–015

Consequence ID	Falsifiable statement	Assumptions
`SCI-FLT-MATCHED-PRED-019`	With two nonzero template amplitudes, the expectation at p equals $R(p, s_1)A_{s_1} + R(p, s_2)A_{s_2}$ plus any nuisance term.	003–007
`SCI-FLT-MATCHED-PRED-020`	A nonzero constant or gradient background changes $\mathrm{E}[\widehat{a}_p(M)]$ by exactly $L_{p,g}b$ when that contraction is nonzero.	003–007
`SCI-FLT-MATCHED-PRED-021`	A nonzero off-diagonal $R_{\text{fixed}}(p, s)$ produces a response at p to a unit template at $s \neq p$ despite exact unit diagonal response.	003–006
`SCI-FLT-MATCHED-PRED-022`	A mismatched signal shape returns its response-weighted template amplitude and need not equal any unconstrained source property.	003–006
`SCI-FLT-MATCHED-PRED-023`	For parent-learned state, an authorized full-procedure response can differ from fixed-state response exactly through the recorded state change; without rerun authority the former remains unavailable.	010
`SCI-FLT-MATCHED-PRED-024`	The operational realized U1 covariance is $\mathrm{Cov}[P_C F_g(M) \mid h_{\text{pre}}]$ on its fixed finite codomain, not the reference covariance. If fixed-state linearity $F_g(M) = \widetilde{L}_g M$ is established, this equals the matrix expression in Equation 12.	001, 002, 009, 013
`SCI-FLT-MATCHED-PRED-025`	If a coordinate in $\mathcal{D}_{\text{loc}}(p)$ has complete authorized template/model/covariance construction authority, lacks observed m_{obs} payload, and has exact-zero final coefficient c_{pq}, the amplitude remains defined and unchanged without dereference. Missing model or covariance authority on that required local coordinate makes `SCI-FLT-MATCHED-AO-001-A`, its optimality claim, and $v_{\text{GLS,reference}} = d_p^{-1}$ unavailable. Changing the coefficient to exact nonzero activates the observed-payload dependency and makes the anchor unavailable when the payload is absent.	004–006,008

10 Unselected authored option sets

All alternatives below remain unselected in this preflight. The package may later authorize more than one separately named route, but every request and realization binds exactly one with no mixing, target-data-driven choice, or fallback. Every scientific weight is exact, self-adjoint, positive semidefinite, and used through exact normalization. Observation and coadd authorizations, selections, and generations remain separate.

10.1 `SCI-FLT-MATCHED-AO-001`: Scientific weighting authority

Each route binds the map chart, units, support, population, Learn/Resolve identity, exact subspaces, window/boundary action, regularization, coefficient roles, diagnostics, state uncertainty, generations, and failure disposition.

10.1.1 `SCI-FLT-MATCHED-AO-001-A`: exact constrained local inverse-covariance GLS

Science.
Declare $\mathcal{D}_{\mathrm{loc}}(p)$ and E_p, restrict exact authorized C_{parent} to that domain, then use $W_p = U_p C_{p,E}^{-1} U_p^{\mathsf{T}}$ on the declared estimable subspace with identifiable projected template. This alone supports the theorem in Section 6.3.
Failure and evidence.
Missing covariance authority, population or conditioning mismatch, nonpositive $C_{p,E}$, subspace mismatch, or unidentifiable template makes GLS/optimality unavailable. Future assessment checks the exact theorem and the preregistered numerical profile separately.

10.1.2 `SCI-FLT-MATCHED-AO-001-B`: structured covariance-derived weighting

Science.
This is an authorship envelope, not yet a selectable weight. A decision-ready successor must supply one concrete exact self-adjoint PSD W_p derived from a declared covariance family. Once authored, that W_p is exact science even when it is not the exact covariance inverse. Normalization and unit response remain; optimality and d_p^{-1} variance do not.
Failure and evidence.
Unbound structure, population, modes, support, regularization, or covariance ancestry fails the route. Any comparison with an inverse covariance is an engineering/profile observable, not an implied claim.

10.1.3 `SCI-FLT-MATCHED-AO-001-C`: radially symmetrized field-power spectral weighting

Identity.
If owner-authorized, the separately named nonoptimal method is `radially_symmetrized_field_power_spectral_weighting` and carries `optimality_status=not_claimed`.
Object.
For each predeclared $\mathcal{D}_{\mathrm{loc}}(p)$, let ξ_q be the declared local-chart coordinate and define the exact analysis operator

$$(A_p u)_k = \alpha_p \sum_{q \in \mathcal{D}_{\mathrm{loc}}(p)} H_p(q) u_q e^{-2\pi i k \cdot \xi_q} \mu_q, \qquad F_{jp}(k) = (A_p E_p f_j)_k,$$

for a declared field population f_j. Define

$$B_{pb} = \{k : \rho_p(k) \in [\rho_{pb}, \rho_{p,b+1})\}, \qquad \overline{P}_{pb} = \frac{\sum_{j,k \in B_{pb}} w_{pjk} |F_{jp}(k)|^2}{\sum_{j,k \in B_{pb}} w_{pjk}}.$$

where $\rho_p(k)$ is the exact radial coordinate under the declared local chart and WCS metric. The source may be an independently authorized noise population or a declared parent-derived field. Its name and deterministic-residual imprint are retained.
Exact conventions.
Each local radial bin B_{pb} is half-open in the declared radial coordinate except the declared final-boundary rule; boundary ties use one deterministic assignment. Member/mode weights are finite and nonnegative and their denominator is positive. Phase, normalization, units, conjugate modes, multiplicities, averaging order, WCS metric, window, and support are exact.

Weight and modes.
Let D_p be diagonal on a conjugate-complete mode set, with

$$(D_p)_{kk} = w_{\mathrm{spec,p}}(k) = \left[\overline{P}_{p,b(k)} + \lambda_{p,b(k)}\right]^{-1}$$

on admitted modes and zero action on explicitly excluded modes, where $\lambda_{pb} \geq 0$ has the units of $\overline{P}_{pb}$. The exact coordinate-basis weight is

$$W_p = A_p^{\dagger} D_p A_p.$$

The selected phase, conjugacy, multiplicity, and real-coordinate restriction shall make W_p real, self-adjoint, and positive semidefinite and shall declare its units and final support. Regularized modes and excluded-null modes are distinct. A sparse bin is never interpolated or borrowed from neighbors without a separately authored rule.
Consequences.
Radialization deliberately loses angular information and implies no noise, covariance, stationarity, isotropy, or optimality claim. d_p is normalization. Matching covariance, if any, requires independent authority.
Failure and evidence.
Invalid weights/denominator, nonfinite or negative power, unbound ties/multiplicity, missing field provenance, incomplete support, or hidden borrowing fails the route. Future evidence reports bin membership, source imprint, anisotropy/dispersion, leakage, regularization, response, and NOI parity as mandatory diagnostics without promoting them to scientific thresholds. Their exact diagnostic estimators and normalizations are preregistered engineering evidence fields and never determine v0.1 admission.

10.1.4 `SCI-FLT-MATCHED-AO-001-D`: declared weaker positive-semidefinite weighting

Science.
This is a successor-authorship trigger, not a selectable catch-all. A future candidate must supply one concrete exact self-adjoint PSD map- or transform-domain W_p with declared purpose, population, units, support, rank/null, regularization, and lineage, but no covariance ancestry claim.
Failure and evidence.
Non-self-adjointness, indefiniteness, hidden coefficients, state drift, incomplete support, or invalid d_p fails the route. No optimality or variance meaning follows.

10.2 `SCI-FLT-MATCHED-AO-002`: Operator realization and numerical-conformance class

No numerical constant in this family is scientific authority. Exactly one engineering profile is preregistered per realized numerical identity.

10.2.1 `SCI-FLT-MATCHED-AO-002-A`: numerical realization of the exact operator

The scientific operator, response, support, subspaces, and failure semantics are exact. The preregistered engineering profile binds floating-point format, algorithms, environment, exact comparator, domains, norms, scales, projectors, coverage, and acceptance rules before results are seen. Profile pass is only a numerical-conformance statement.

10.2.2 `SCI-FLT-MATCHED-AO-002-B`: intentionally distinct scientific operator

This label is a nonselectable successor-authorship trigger. A concrete route is admissible only through separate owner authority and authorship that states the affected quantity, estimand, response/support/null discrepancy, scientific error budget, norm/domain, calibration consequence, consumers, and rationale. It creates a distinct method or realization identity; it is not ordinary finite precision and cannot borrow the exact operator's response/covariance.

10.2.3 `SCI-FLT-MATCHED-AO-002-C`: numerical route unavailable

If no preregistered profile, exact comparator, domain/coverage, or required projector exists, no numerical-conformance claim is available. No threshold, iteration count, percentile, or post-hoc test supplies a fallback.

10.3 SCI-FLT-MATCHED-AO-003: Conditional-covariance scope and representation

This family requires one scientific-scope disposition from A–C and, for an available scope, one exact-representation disposition from D–E. Equivalent exact representations retain one covariance identity. Every signal bundle retains the identity and lifecycle status of each named covariance role, including an unavailable role.

10.3.1 SCI-FLT-MATCHED-AO-003-A: complete named-role covariance scope

Every authorized entry and correlation of the actual produced field is in scope on the full valid output domain for one explicitly named covariance role, with exact random variables, population, conditioning, units, rank/null, response, and generation. U1 conditional stochastic covariance is not a joint U1+U2 calibration/template product by implication.

10.3.2 SCI-FLT-MATCHED-AO-003-B: named projected scientific covariance scope

Only $P_c C^{(r)}_{U1,\mathrm{realized}} P_c^{\mathsf{T}}$ for an explicitly named covariance role r and consumer projection is in scope. The role, projection basis, measure, rank, validity, and consumer are exact; all outside correlations remain unknown.

10.3.3 SCI-FLT-MATCHED-AO-003-C: covariance unavailable

Missing parent-covariance or cross-term authority is typed unavailability. The signal may remain complete, but no operational covariance, covariance-based standardization, significance, draw, independence, or covariance-dependent use follows. A separately theorem-supported $v_{\mathrm{GLS,reference}}(p) = d_p^{-1}$ under SCI-FLT-MATCHED-AO-001-A may remain available. This typed companion status does not make the signal bundle partial unless an exact named-use policy requires the covariance companion.

10.3.4 SCI-FLT-MATCHED-AO-003-D: exact resident covariance representation

Represent the selected scope by an explicit array or exact structured object. The representation type and query vocabulary are declared; every supported entry/quadratic form is exact, and no representation-only change alters the scientific covariance identity.

10.3.5 SCI-FLT-MATCHED-AO-003-E: exact lineage/on-demand covariance representation

Persist exact immutable lineage sufficient to reconstruct every supported query without undocumented state. Each result is immutable and carries the same covariance identity. Missing lineage or unsupported queries are unavailable, never zero.

10.4 SCI-FLT-MATCHED-AO-004: Immutable-state representation

The scientific invariant for every alternative is exact reproducibility of the resolved immutable state and its required query vocabulary. Representation alone does not change the estimator or signal generation. After the scientific query vocabulary is fixed, A–C are engineering representation selections.

10.4.1 SCI-FLT-MATCHED-AO-004-A: full materialization

Persist every resolved state element with exact indexing, units, population, generation, and canonical identity. Reload shall reproduce every supported operator/state query exactly under the numerical profile.

10.4.2 SCI-FLT-MATCHED-AO-004-B: compact exact state

Persist an exact factorization or sufficient-state representation plus complete query semantics. Discarded information is allowed only when proved irrelevant to every supported query. Unsupported queries are unavailable.

10.4.3 SCI-FLT-MATCHED-AO-004-C: exact lineage reconstruction

Persist immutable inputs, Learn/Resolve algorithms and versions, environment facts, and identity needed for deterministic reconstruction. Missing or mutable lineage fails dependent routes; relearning is not a fallback.

10.5 SCI-FLT-MATCHED-AO-005: Response science and exact representation

Before representation, every route binds fixed/FP/realized/reference type, anchor domains, index/measure, query vocabulary, validity, units/calibration, state/operator/template generation, consumer scope, retention, and failure. FLT does not guess a future FRUIT query. After the scientific response type/domain/query vocabulary is fixed, A–C are engineering representation selections.

10.5.1 SCI-FLT-MATCHED-AO-005-A: full response materialization

Materialize every response entry in the authorized domain with exact index maps and identity. A matrix does not by itself authorize a beam, moment, integral, or other reduced interpretation.

10.5.2 SCI-FLT-MATCHED-AO-005-B: exact structured response

Persist an exact factorization or mathematical object that answers the entire declared query vocabulary without unproved invariance. Unsupported queries are typed unavailable.

10.5.3 SCI-FLT-MATCHED-AO-005-C: exact lineage/on-demand response

Persist complete immutable lineage needed to reconstruct supported response queries without undocumented state. Query results are immutable; failure does not substitute a reduced kernel.

10.6 SCI-FLT-MATCHED-AO-006: Role-separated SCI-VAL verdict representation

The normative policy is one dependency graph over seven separate verdicts: PA, SA, SP, CU, NU, RU, and FH. Each retains request, applicability, eligibility, realization, producer facts, policy, action, lifecycle, evidence, and version. The alternatives are lossless engineering representations of the same owner-disposed policy and do not require a scientific-owner selection.

10.6.1 SCI-FLT-MATCHED-AO-006-A: seven separate records

Persist one immutable record per role. A view may list all verdicts but has no actionable scalar aggregate.

10.6.2 SCI-FLT-MATCHED-AO-006-B: grouped layered records

Group roles into declared layers only while retaining seven independently addressable subverdicts and the full dependency graph. A layer summary cannot replace the subverdict vector.

10.6.3 SCI-FLT-MATCHED-AO-006-C: one seven-role verdict vector

Persist one composite with seven separately addressable complete records. Consumers request a named component. A scalar may be human-readable only and shall never drive action.

10.7 Option closure

Until exact dispositions exist, affected weighting, numerical-conformance, covariance, state-query, response-query, SCI-VAL, and handoff routes remain unavailable. Scientific-owner dispositions choose scientific meanings; engineering preregistration/representation declarations choose exact realization form only after those meanings are fixed;

incomplete catch-alls trigger successor authorship. No disposition establishes implementation conformity, response or covariance fidelity, observational validation, performance, readiness, production, freeze, or Unity status.

11 Edge cases, exclusions, and failure routing

Condition	Scientific meaning	Required handling
Empty or incomplete application support	No defined reference estimand at the affected anchor	Mark `unavailable` before evaluation; no padding, extension, imputation, partial estimate, or renormalization.
Numerical fill near an edge	Merely a computational placeholder	Admit an output only after exact erosion proves zero dependency on every filled value.
Missing payload on a construction-only row with exact-zero c_{pq}	The row is in $\mathcal{D}_{\mathrm{loc}}(p)$ but not $\mathcal{S}_{\mathrm{apply}}(p)$	Do not dereference the observed payload only when exact template/model/covariance construction authority exists; the amplitude remains defined and unchanged. If $c_{pq} \neq 0$, the row becomes an active dependency and the anchor is Unavailable. No coefficient threshold is permitted.
Missing stochastic-model or covariance authority on a required local row	`SCI-FLT-MATCHED-AO-001-A` has no complete local law even if the final application coefficient is zero	Mark `SCI-FLT-MATCHED-AO-001-A`, its optimality claim, and $v_{\mathrm{GLS,reference}} = d_p^{-1}$ `unavailable`. Evaluate a sparse non-GLS route only when separately authorized and exactly requested; no fallback.
$d_p = 0$, $d_p < 0$, nonfinite d_p, or complex d_p	Template is null, weighting is invalid, or numerical identity failed	Typed unavailability or attempt failure; never publish zero amplitude, infinite variance, or fallback.
Template lies partly in $\mathcal{N}(W_p)$	Some declared template modes are unestimable	Allowed only if the selected option explicitly defines the same estimable subspace and $d_p > 0$; record response loss and null space. Otherwise unavailable.
Indefinite or non-self-adjoint W_p	Weight is outside the admitted scientific bilinear family semantics	Fail selected `SCI-FLT-MATCHED-AO-001` closure unless an exact distinct operator and inner product were separately authored; no silent symmetrization.
Covariance zero-variance mode	Parent law declares deterministic variation in that direction	Keep separate from excluded, weighting-null, and template- unidentifiable modes. It is outside the v0.1 positive-definite-subspace theorem unless separately authored; do not broaden the stated optimum.
Excluded or infinite-variance mode	Policy deliberately removes a direction from the estimable subspace	Record the exact projector and response loss; do not call it a covariance zero mode.
Weighting-null mode	W_p assigns zero action to a direction	Record the weighting null space; no covariance meaning follows without separate authority.
Projected template is zero	The requested amplitude is unidentifiable in the declared subspace	Anchor unavailable; no Moore–Penrose or normalization fallback.

Condition	Scientific meaning	Required handling
Zero, negative, or nonfinite field-power bin	No finite declared reciprocal weighting for that mode/bin	Apply only owner-declared regularization within the frozen state; otherwise exclude as a declared null mode or make route unavailable. Never infer infinite precision.
Sparse radial bin or large angular scatter	Radially averaged PSD lacks declared information or radial sufficiency	Report the preregistered effective-sample, anisotropy/dispersion, and leakage diagnostics. They are not v0.1 admission predicates and have no owner-set failure thresholds. Exact weight/denominator/support predicates still apply; no neighboring-bin interpolation unless separately authored.
Unknown covariance entry	Scientific knowledge is absent	Preserve unknown; do not encode zero, diagonal independence, or a usable inverse.
Available W but unavailable C_{parent}	Weighting exists without covariance authority	Amplitude filtering may proceed under a selected weaker option; optimality, reference variance, and operational covariance are unavailable.
Exact GLS premises with operational covariance unavailable	Reference theorem and operational product have different roles	Preserve $v_{\text{GLS,reference}} = d_p^{-1}$ when independently established; mark the operational covariance role Unavailable and forbid covariance-derived uses.
Available C_{parent} with mismatched population/state	Covariance does not describe this realization	Covariance route unavailable; no rescaling or population substitution.
Template rescaled by $c \neq 0$	Amplitude convention changes	Create a new template/realization identity. The $\widehat{a} \mapsto \widehat{a}/c$ law holds only for a pure amplitude-coordinate reparameterization with $\mathcal{D}_{\text{loc}}$, E_p, W_p, g, support, and non-template state fixed. Any re-Learn or re-Resolve operation is a new realization outside that law; units and lineage must follow in every case.
Template phase or anchor changed	Scientific response convention changes	New template and realization; never treat as a numeric implementation detail.
Template mismatch to a real signal	Base matching-template estimand does not equal an unconstrained source property	Report only response-weighted shape amplitude; no bias correction, morphology, or new uncertainty without separate authority.
Parent beam or calibration exists	Provenance is present, not inherited matched-filter meaning	Preserve lineage; require exact authority before flux, matched beam/solid angle, extended-source, or calibration-covariance claim.
Coadd membership changes	Parent identity changes	New coadd parent, state, realization, response, and products; do not reuse observation/coadd state by implication.

Condition	Scientific meaning	Required handling
NOI member fails K_{NOI} or suggests a new state	Admission or per-member relearning would change the ensemble method	Mark the member or affected anchor incompatible/unavailable before Apply. A learned update creates a successor generation; no support adaptation or result-dependent admission.
Approximation misses one null or support mode	Estimand/validity differs even if aggregate error is small	Fail the preregistered profile; if intentional, assign a distinct scientific identity. No percentile hides the disagreement.
Iteration/tail cap reached	Work limit, not convergence evidence	Accept only if every selected quantitative envelope bound independently passes; otherwise fail.
Optional companion fails after signal completion	Companion atomicity failed, not necessarily signal science	Mark companion failed; preserve signal complete publication candidate unless exact named-use policy made the companion required.
Complete candidate requested before realization	Product closure is out of order and required immutable values do not yet exist	Fail product closure; do not evaluate publication policy until a realized candidate is complete.
Optional role not requested	No child or companion was requested	Record `not_requested`; do not narrow $\mathcal{V}_{\mathrm{signal}}$ or block another role.
Normative per-anchor unavailability	The validity rule defined no amplitude at that anchor	Preserve the unavailable status inside an otherwise role-complete field; this does not by itself make the bundle partial.
Required signal bundle member fails	A required field, mask, identity, response reference, lifecycle record, or lineage member is absent	Fail the entire signal bundle; do not disguise the missing member as unavailable pixels inside an otherwise complete product.
Additional method route is package-authorized	Package authorization is not a per-realization selection	Preserve each separately named route. Require one exact requested method identity; never mix, choose from target data, or fall back between routes.
Successor state or option is approved	Scientific identity advances	Create immutable successor generation and mark predecessor superseded only by policy; never mutate predecessor facts.
Unselected or unavailable option requested	No authorized route exists	Fail closed and name the missing owner disposition; no automatic selector or fallback.
Response query is unsupported	Scientific interface does not define that quantity	Return typed unavailability; never approximate with a kernel, radial profile, parent beam, or zero.
Fixed-state response requested as full-procedure response	State-learning dependence has not been rerun	Return full-procedure response unavailable and retain the fixed-state type; no consumer may strengthen it.
Reference response or covariance attached to a realized field	Comparator is being mislabeled as the actual product consequence	Fail the companion or bundle policy; attach the operational realized response and named-role $C_{U1,\mathrm{realized}}$ when authorized.

Condition	Scientific meaning	Required handling
Failure-bearing operational covariance sample	$P_C F_g(M)$ is not a fixed- dimensional finite numerical random vector almost surely on the predeclared domain	Mark the covariance role `unavailable`. Do not condition on success, censor, pairwise-delete, or change domains without a separately named population and covariance role; signal status is unchanged unless named-use policy requires the companion.
FLT to FRUIT consumer asks outside $\mathcal{Q}^{0.1}_{\mathrm{FLT}}$ or for FRUIT behavior	Request crosses the current envelope or package authority	Return typed unavailability and require a successor FLT envelope or separate FRUIT contract as applicable; existing FLT signal and handoff facts remain unchanged.
Filtered-map peak is examined	A numerical feature is not an authorized source object	No detection, candidate, significance, completeness, purity, fit, morphology, catalog, or SRC claim follows.
Historical title or filename contains "optimal"	Human-facing terminology is not theorem evidence	Require explicit `optimality_status`; under `SCI-FLT-MATCHED-AO-001-C` it is always `not_claimed`.

11.1 Explicit method exclusions

Posterior/Wiener sky reconstruction, generic ordinary-convolution semantics, source or candidate learning, data-thresholded destriping, automatic selectors/fallbacks, adaptive edge/background behavior, the historical high-pass/delta case, partial-support estimation, FLT-owned coaddition, source analysis, and FRUIT algorithms are not authorized by this contract. A proved special-case numerical coincidence with normalized correlation/convolution retains the `SCI-FLT-MATCHED` estimand and does not authorize generic convolution semantics. Other methods require separate scientific contracts if later commissioned.

`SCI-FLT-FIXED` remains a separate protected package. This contract neither selects, modifies, supersedes, nor supplies scientific authority for that method.

The historical compatibility term "Wiener filter" may refer to this method only when accompanied by the exact `SCI-FLT-MATCHED` identity and may not imply a sky prior, posterior estimand, or Wiener reconstruction.

11.2 Nonclaim closure

Passing a scientific contract consequence or an engineering conformance test establishes only that named claim for the exact immutable realization. Validation-profile pass is not achieved performance. Achieved performance is not readiness. Readiness is not production. None of those layers can select an authored option, repair missing scientific authority, or broaden this validity domain.

Part III

Engineering audit protocol

12 Conformance test families

These are specification-level evidence families, not executed tests. Each future realization shall instantiate all applicable families after the exact scientific, representation, and numerical-profile dispositions exist.

ID	Scope	Minimum evidence and verdict rule
`SCI-FLT-MATCHED-CT-001`	Identity and authority closure	Freeze contract, realization, selected AO, parent/template/state, units, validity, lifecycle, and provenance identities. Any missing/mutable identity is unavailable; mismatch fails.
`SCI-FLT-MATCHED-CT-002`	Parent admission	Exercise observation and coadd admissions separately; reject derived/JINC parents, missing coadd membership, changed generation, and assumed commutation/state sharing.
`SCI-FLT-MATCHED-CT-003`	Anchors, template, and units	Verify the identical parent-pixel anchor lattice, deterministic phase/ties, exact template object and query representation, scaling, lineage, $[\widehat{a}] = [m_{\mathrm{obs}}]/[t]$, rescaling law, and rejection of target/member/candidate/output/residual/post-selection template choice or unsupported flux/beam meaning.
`SCI-FLT-MATCHED-CT-004`	Exact operator and normalization	Compare $\mathcal{S}_{\text{parent fact}}$, $\mathcal{D}_{\text{model}}$, M, $\mathcal{D}_m$, m_{obs}, $\mathcal{D}_{\text{loc}}(p)$, E_p, t_p, d_p, $\ell^\star_p$, c_p, $\mathcal{S}_{\text{apply}}$, sparse amplitude, and L, local restriction/inversion, and matching-template response with an independent exact oracle; test self-adjoint PSD closure and every invalid-d_p route.
`SCI-FLT-MATCHED-CT-005`	Support, boundaries, and subspaces	Construct predeclared local domains and complete, eroded, filled, missing, invalid, final-zero-coefficient, and each distinct covariance/excluded/weighting/template-null case; require exact domain, support-family, causal-lineage, and projector equality with no partial estimate or edge renormalization. Explicitly verify that a missing construction-only observed payload with exact-zero coefficient is never dereferenced when complete construction authority exists; missing model/covariance authority makes `SCI-FLT-MATCHED-AO-001-A` unavailable; and a nonzero coefficient makes the anchor unavailable when observed payload is absent.
`SCI-FLT-MATCHED-CT-006`	Weighting option	Instantiate the selected `SCI-FLT-MATCHED-AO-001` population, units, coefficients, rank/null, regularization, window/edge, response, uncertainty, products, failure, and option-specific exact closure. Verify package-, parent-class-, and named-use authorization separately from the one exact method requested by each realization, with no mixing, target-data selection, or fallback. Test observation and coadd separately. For `SCI-FLT-MATCHED-AO-001-C`, report the preregistered effective-sample, anisotropy/dispersion, and leakage diagnostics without treating them as v0.1 admission thresholds, and require `optimality_status=not_claimed`.

ID	Scope	Minimum evidence and verdict rule
`SCI-FLT-MATCHED-CT-007`	Numerical conformance	Evaluate F_g under the one preregistered profile over every declared state, support, boundary, subspace, response, and covariance stratum. Keep realized and reference quantities separate; report complete observables and coverage; use e_d only when the profile declares a separable numerical normalization $\widetilde{d}_p$; use linear-operator comparisons only after linearity is established, without promoting thresholds to scientific authority.
`SCI-FLT-MATCHED-CT-008`	Response	Verify fixed, full-procedure, realized, and reference types; unit self-response; general off-diagonal behavior; selected `SCI-FLT-MATCHED-AO-005` domain/query/representation contract; lineage; lifecycle; and unsupported-query rejection. Directional-derivative and finite-difference evidence shall bind observed reference parent m_{obs}, step, sidedness, and one common finite numerical query domain.
`SCI-FLT-MATCHED-CT-009`	State and NOI parity	Verify Learn–Resolve–Apply inputs, population, protection, diagnostics, freeze, selected `SCI-FLT-MATCHED-AO-004` exact query reproduction, generation immutability, successor behavior, pre-Apply K_{NOI} admission, identical admitted-member transformation, and rejection of result-dependent admission, adapted support, or per-member relearning.
`SCI-FLT-MATCHED-CT-010`	Covariance and uncertainty	Where authorized, evaluate operational $C_{U1,\mathrm{realized}} = \mathrm{Cov}[P_C F_g(M) \mid h_{\mathrm{pre}}]$ and exact reference covariance separately; assess the fixed finite codomain, almost-sure numerical-production premise, each selected `SCI-FLT-MATCHED-AO-003` named-role scope and exact representation, units, random variables, population, conditioning, subspaces, consumers, and any separately authorized U1+U2 joint product. Verify that h_{pre} excludes numerical m_{obs} values/digests, success, draw-dependent identities/domains, publication, censoring, and pairwise deletion. Empirical evidence follows the preregistered sampling protocol above. Otherwise verify typed role/status unavailability, unchanged signal completeness unless a named-use policy requires the companion, and the independent availability or unavailability of $v_{\mathrm{GLS,reference}} = d_p^{-1}$ under exact `SCI-FLT-MATCHED-AO-001-A`.
`SCI-FLT-MATCHED-CT-011`	Products and lifecycle	Verify atomic signal membership, companion independence, named covariance-role identity/status in every signal bundle, the complete lifecycle vocabulary and exact Applied–Realized–Complete–PublicationDecided–Published/NotProduced order, required/optional failure propagation, per-role validity domains, immutability, supersession, no public unqualified intermediate, and exact lineage closure.

ID	Scope	Minimum evidence and verdict rule
SCI-FLT-MATCHED-CT-012	SCI-VAL profiles and actions	First reject the returned role- semantics drafts as unregistered and unevaluable. For an exact later owner-approved Registry record, instantiate the engineering-declared `SCI-FLT-MATCHED-AO-006` representation of the owner-disposed semantics; exhaust four-axis verdict/dependency/action combinations for PA, SA, SP, CU, NU, RU, and FH; verify lossless layout, aggregation nonconcealment, separate request/status addressability, unrequested optional role nonblocking, and VAL's evaluator-only role.
SCI-FLT-MATCHED-CT-013	Boundaries and FLT-to-FRUIT envelope	Verify exact MAP, template, NOI, and one-way FRUIT producer-envelope records; K_{NOI}; finite $\mathcal{Q}^{0.1}_{\mathrm{FLT}}$; route status; selected `SCI-FLT-MATCHED-AO-004` state and `SCI-FLT-MATCHED-AO-005` response reconstruction consequences; and all supported queries without undocumented state. Exercise all six NOI/FH request states and reject FRUIT science or unsupported queries without changing the signal verdict.
SCI-FLT-MATCHED-CT-014	Exclusions and fail-closed selection	Attempt every excluded or unselected route, automatic fallback, source interpretation, posterior/Wiener meaning, unauthorized generic-convolution semantics, and unsupported uncertainty claim; require typed rejection with no substituted product.
SCI-FLT-MATCHED-CT-015	Claim-layer separation	Verify records distinguish scientific authority, conformity, representation fidelity, SCI-VAL evaluation, observational validation, achieved performance, readiness, and production, and that no transition is inferred from another layer.

13 Engineering representation-selection rationale

These comparisons do not select science. They become actionable only after the scientific role and finite query vocabulary are fixed and exact equivalence is demonstrated.

Choice	Storage and query	Reproduction dependencies	Failure surface	Audit burden
`SCI-FLT-MATCHED-AO-003-D`, `SCI-FLT-MATCHED-AO-004-A`, `SCI-FLT-MATCHED-AO-005-A`	Highest resident storage; direct supported queries	Lowest reconstruction dependence; immutable object reader still required	Corruption, indexing, unit, or schema mismatch	Byte/object integrity plus exhaustive query checks.

Choice	Storage and query	Reproduction dependencies	Failure surface	Audit burden
SCI-FLT-MATCHED-AO-004-B, SCI-FLT-MATCHED-AO-005-B and structured SCI-FLT-MATCHED-AO-003-D	Lower storage; queries require exact factorization/object evaluation	Exact structure, algorithm identity, and query semantics	Hidden approximation, unsupported query, rank/phase loss, or noncanonical factorization	Prove equivalence for the complete finite vocabulary and audit factor identity.
SCI-FLT-MATCHED-AO-003-E, SCI-FLT-MATCHED-AO-004-C, SCI-FLT-MATCHED-AO-005-C	Lowest resident result storage; potentially highest query latency	Complete immutable lineage, algorithms, versions, environment facts, and deterministic reconstruction	Missing or mutable lineage, nonreproducible environment, or reconstruction failure	Reconstruct representative and boundary queries and verify lineage closure.
SCI-FLT-MATCHED-AO-006-A	Largest verdict-record count; direct role lookup	Seven independent records and dependency graph	Cross-record version drift or incomplete role set	Audit role completeness and graph consistency.
SCI-FLT-MATCHED-AO-006-B	Moderate record count; layered navigation	Lossless subverdict addressing within each layer	Layer summary conceals or overrides a subverdict	Audit every subverdict and aggregation nonauthority.
SCI-FLT-MATCHED-AO-006-C	Compact access to one seven-role object	Stable component indexing and complete embedded records	Scalarization or component omission	Audit all seven components and forbid actionable scalar use.

14 Requirement-to-test routing

Test family	Primary normative coverage
SCI-FLT-MATCHED-CT-001	SCI-FLT-MATCHED-REQ-001, SCI-FLT-MATCHED-REQ-007, SCI-FLT-MATCHED-REQ-013, SCI-FLT-MATCHED-REQ-032, SCI-FLT-MATCHED-REQ-034, SCI-FLT-MATCHED-REQ-038, SCI-FLT-MATCHED-REQ-039, SCI-FLT-MATCHED-REQ-048.
SCI-FLT-MATCHED-CT-002	SCI-FLT-MATCHED-REQ-003, SCI-FLT-MATCHED-REQ-004, SCI-FLT-MATCHED-REQ-016, SCI-FLT-MATCHED-REQ-037, SCI-FLT-MATCHED-REQ-040; SCI-FLT-MATCHED-PRED-006.
SCI-FLT-MATCHED-CT-003	SCI-FLT-MATCHED-REQ-002, SCI-FLT-MATCHED-REQ-005, SCI-FLT-MATCHED-REQ-006, SCI-FLT-MATCHED-REQ-007, SCI-FLT-MATCHED-REQ-036, SCI-FLT-MATCHED-REQ-040; SCI-FLT-MATCHED-PRED-002, SCI-FLT-MATCHED-PRED-003, SCI-FLT-MATCHED-PRED-014, SCI-FLT-MATCHED-PRED-022.
SCI-FLT-MATCHED-CT-004	SCI-FLT-MATCHED-REQ-008, SCI-FLT-MATCHED-REQ-009, SCI-FLT-MATCHED-REQ-012, SCI-FLT-MATCHED-REQ-041, SCI-FLT-MATCHED-REQ-043, SCI-FLT-MATCHED-REQ-050; SCI-FLT-MATCHED-PRED-001–SCI-FLT-MATCHED-PRED-005, SCI-FLT-MATCHED-PRED-019–SCI-FLT-MATCHED-PRED-021.
SCI-FLT-MATCHED-CT-005	SCI-FLT-MATCHED-REQ-007, SCI-FLT-MATCHED-REQ-009, SCI-FLT-MATCHED-REQ-010, SCI-FLT-MATCHED-REQ-033, SCI-FLT-MATCHED-REQ-042, SCI-FLT-MATCHED-REQ-050; SCI-FLT-MATCHED-PRED-004, SCI-FLT-MATCHED-PRED-005, SCI-FLT-MATCHED-PRED-010, SCI-FLT-MATCHED-PRED-025.

Test family	Primary normative coverage
SCI-FLT-MATCHED-CT-006	SCI-FLT-MATCHED-REQ-012–SCI-FLT-MATCHED-REQ-014, SCI-FLT-MATCHED-REQ-019, SCI-FLT-MATCHED-REQ-021; all SCI-FLT-MATCHED-AO-001 consequences.
SCI-FLT-MATCHED-CT-007	SCI-FLT-MATCHED-REQ-008, SCI-FLT-MATCHED-REQ-011, SCI-FLT-MATCHED-REQ-033, SCI-FLT-MATCHED-REQ-045; SCI-FLT-MATCHED-PRED-010, SCI-FLT-MATCHED-PRED-011, SCI-FLT-MATCHED-PRED-024; all SCI-FLT-MATCHED-AO-002 consequences.
SCI-FLT-MATCHED-CT-008	SCI-FLT-MATCHED-REQ-005, SCI-FLT-MATCHED-REQ-011, SCI-FLT-MATCHED-REQ-028, SCI-FLT-MATCHED-REQ-036, SCI-FLT-MATCHED-REQ-043–SCI-FLT-MATCHED-REQ-045, SCI-FLT-MATCHED-REQ-048; SCI-FLT-MATCHED-PRED-001, SCI-FLT-MATCHED-PRED-015, SCI-FLT-MATCHED-PRED-017, SCI-FLT-MATCHED-PRED-019–SCI-FLT-MATCHED-PRED-023; all SCI-FLT-MATCHED-AO-005 consequences.
SCI-FLT-MATCHED-CT-009	SCI-FLT-MATCHED-REQ-015–SCI-FLT-MATCHED-REQ-018, SCI-FLT-MATCHED-REQ-038, SCI-FLT-MATCHED-REQ-042, SCI-FLT-MATCHED-REQ-044, SCI-FLT-MATCHED-REQ-047; SCI-FLT-MATCHED-PRED-006, SCI-FLT-MATCHED-PRED-007, SCI-FLT-MATCHED-PRED-011, SCI-FLT-MATCHED-PRED-023; all SCI-FLT-MATCHED-AO-004 consequences.
SCI-FLT-MATCHED-CT-010	SCI-FLT-MATCHED-REQ-020–SCI-FLT-MATCHED-REQ-022, SCI-FLT-MATCHED-REQ-025, SCI-FLT-MATCHED-REQ-036, SCI-FLT-MATCHED-REQ-045, SCI-FLT-MATCHED-REQ-046; SCI-FLT-MATCHED-PRED-008, SCI-FLT-MATCHED-PRED-009, SCI-FLT-MATCHED-PRED-013, SCI-FLT-MATCHED-PRED-024; all SCI-FLT-MATCHED-AO-003 consequences.
SCI-FLT-MATCHED-CT-011	SCI-FLT-MATCHED-REQ-023–SCI-FLT-MATCHED-REQ-027, SCI-FLT-MATCHED-REQ-038; SCI-FLT-MATCHED-PRED-011–SCI-FLT-MATCHED-PRED-013.
SCI-FLT-MATCHED-CT-012	SCI-FLT-MATCHED-REQ-019, SCI-FLT-MATCHED-REQ-026, SCI-FLT-MATCHED-REQ-032, SCI-FLT-MATCHED-REQ-035, SCI-FLT-MATCHED-REQ-049; SCI-FLT-MATCHED-PRED-016; all SCI-FLT-MATCHED-AO-006 consequences.
SCI-FLT-MATCHED-CT-013	SCI-FLT-MATCHED-REQ-028, SCI-FLT-MATCHED-REQ-029, SCI-FLT-MATCHED-REQ-038, SCI-FLT-MATCHED-REQ-048; SCI-FLT-MATCHED-PRED-017, SCI-FLT-MATCHED-PRED-018; selected SCI-FLT-MATCHED-AO-004 and SCI-FLT-MATCHED-AO-005 handoff consequences.
SCI-FLT-MATCHED-CT-014	SCI-FLT-MATCHED-REQ-006, SCI-FLT-MATCHED-REQ-010, SCI-FLT-MATCHED-REQ-019, SCI-FLT-MATCHED-REQ-027, SCI-FLT-MATCHED-REQ-029–SCI-FLT-MATCHED-REQ-032; SCI-FLT-MATCHED-PRED-004, SCI-FLT-MATCHED-PRED-005, SCI-FLT-MATCHED-PRED-016, SCI-FLT-MATCHED-PRED-018.
SCI-FLT-MATCHED-CT-015	SCI-FLT-MATCHED-REQ-034, SCI-FLT-MATCHED-REQ-039; SCI-FLT-MATCHED-PRED-018.

15 Contract-consequence evidence patterns

Consequence IDs	Evidence pattern	Failure effect
SCI-FLT-MATCHED-PRED-001–SCI-FLT-MATCHED-PRED-003	Exact matching-template injections spanning amplitude, phase/anchor, parent class, state, location, and boundary strata; independent analytic expectation and repeated zero-mean ensemble where needed.	Scientific or operator claim fails.
SCI-FLT-MATCHED-PRED-004–SCI-FLT-MATCHED-PRED-005	Deliberate missing support and each invalid-d_p class; inspect values, validity, lifecycle, errors, and absence of zero/fallback.	Validity or failure contract fails.
SCI-FLT-MATCHED-PRED-006–SCI-FLT-MATCHED-PRED-007	Cross-parent identity comparisons and K_{NOI}-admitted NOI member traces/hashes of state, support, d_p, L, and failure decisions.	State/NOI conformity fails.

Consequence IDs	Evidence pattern	Failure effect
SCI-FLT-MATCHED-PRED-008–SCI-FLT-MATCHED-PRED-010	Independent exact covariance identity, label-removal audit, and full selected-profile metric bundle. Any empirical study separately binds the preregistered sampling protocol in the evidence rules.	Affected covariance or numerical-conformance route fails.
SCI-FLT-MATCHED-PRED-011–SCI-FLT-MATCHED-PRED-013	Mutation/supersession attempts and atomic failure injection for every required and optional member under each named policy.	Identity or product lifecycle fails.
SCI-FLT-MATCHED-PRED-014–SCI-FLT-MATCHED-PRED-017	Pure fixed-state amplitude-coordinate rescaling, separate re-Learn/re-Resolve rescaling, noninvariant response probes, unavailable-route requests, and complete FLT-side handoff query suite.	Unit, response, selection, or handoff route fails.
SCI-FLT-MATCHED-PRED-018	Search every public field/profile/action for excluded semantics or claim-layer promotion; attempt prohibited consumer interpretations.	Conformance claim fails closed.
SCI-FLT-MATCHED-PRED-019–SCI-FLT-MATCHED-PRED-022	Two-template, constant/gradient background, off-diagonal-response, and mismatched-shape constructions compared with Equation 6.	General-sky response or fit boundary fails.
SCI-FLT-MATCHED-PRED-023–SCI-FLT-MATCHED-PRED-024	Authorized Learn–Resolve perturbation reruns, operational F_g response/covariance evaluation, and a fixed-linearity proof before any $\widetilde{L}_g$ matrix oracle, with state changes and actual/reference roles kept distinct.	Response-family or realized-covariance claim fails.
SCI-FLT-MATCHED-PRED-025	Exercise three distinct cases at a local construction coordinate: (1) complete template/model/covariance authority, missing observed payload, and exact-zero final c_{pq} gives no dereference and unchanged amplitude; (2) missing required model or covariance authority makes `SCI-FLT-MATCHED-AO-001-A`, its optimality claim, and reference variance unavailable even when $c_{pq} = 0$; and (3) exact-nonzero c_{pq} with missing observed payload makes the anchor unavailable.	Stochastic-model or numerical application-domain closure fails.

16 Verdict aggregation

A complete implementation-conformity verdict requires every applicable requirement, contract consequence, selected-option consequence, edge-case rule, and named-use action to be complete on the full declared domain. Unavailable is not a pass. Failed is not unavailable. Disabled is not an attempted result. Superseded evidence remains immutable and cannot substantiate a successor unless the successor explicitly binds it and closes every changed fact.

Optional companions receive separate verdicts. A signal-level aggregate may remain complete when an optional companion fails only if the selected FLT-owned policy says that companion is not required. Profile representation follows the selected `SCI-FLT-MATCHED-AO-006` alternative and always retains seven distinct role verdicts.

17 Owner-disposition and engineering-routing guide

Layer	Disposition authority	Conformance consequence
A	Scientific owner, package level	Authorize separately named method routes; recommended A and C may coexist while B/D remain successor triggers.
B	Scientific owner, parent class	Record observation and coadd authorizations separately.
C	Scientific owner, named use	Record allowed routes and required companions for each use.

Layer	Disposition authority	Conformance consequence
D	Request/realization authority	Bind exactly one authorized route; reject mixing, target-data choice, and fallback.
E	Engineering after science	Declare the numerical profile and lossless state, response, covariance, and verdict representations.
F	Typed route state	Preserve unavailable and successor-trigger routes without substitution.

The method ID remains `SCI-FLT-MATCHED`. The recommended owner title is "Matched-template map amplitude estimation"; "Normalized matched-template map filtering" is the second option. Retaining the historical title is a third option only with explicit `optimality_status` on every product and boundary. No title, filename, or ID is evidence. `SCI-FLT-MATCHED-AO-001-A` is the only recommended route eligible for `established_exact_local_GLS`; `SCI-FLT-MATCHED-AO-001-C` and every weaker route carry `not_claimed`. This draft does not select the title for the owner.

Recommended scientific routing is: authorize `SCI-FLT-MATCHED-AO-001-A` as the sole exact local-GLS optimal/reference-variance route when all premises hold; retain B and D as successor-authorship triggers; permit C separately as `radially_symmetrized_field_power_spectral_weighting`, with mandatory diagnostics that are evidence rather than validity thresholds and with no noise, covariance, stationarity, isotropy, or optimality claim. For `SCI-FLT-MATCHED-AO-003`, the owner chooses complete, projected, or unavailable scope per role before engineering chooses resident or lineage representation. The owner fixes the `SCI-FLT-MATCHED-AO-004`/`SCI-FLT-MATCHED-AO-005` finite query vocabularies before representation. The seven `SCI-FLT-MATCHED-AO-006` role meanings and dependency graph remain normative; layout is engineering. Observation and coadd choices remain separate and no fallback is automatic.

A Audit artifact minimums

For each instantiated conformance test, retain: test-family ID; requirement, contract-consequence, AO, assumption, edge-case, and profile IDs; immutable realization and comparator identities; input and expected-output identities; exact units, coordinates, support, state, population, rank/null, and calibration facts; procedure identity; environment-dependent facts; observed results; complete metric arrays or lossless references; tolerances and aggregation; worst-case locations/modes; sampling uncertainty; lifecycle transitions; verdict and reason; and provenance. A summary without these closures is not conformance evidence.

B Author-packet basis and firewall statement

Stage B r0.1 used only the exact eight objects named by the approved author packet. The r0.2 revision additionally used only the scientific owner's exact targeted closure directive dated 2026-08-31. The r0.3 repair used the owner's instruction to apply the independent ChatGPT Pro r0.2 document-review findings; the reviewer verified the two PDF hashes and inspected no implementation. The r0.4 repair used the second independent ChatGPT Pro review of the exact r0.3 PDF pair and the owner's approval of selectable covariance scope plus diagnostics-only AO-001-C admission policy. The r0.5 repair used only the exact scientific-owner directive dated 2026-09-01 and its resulting owner-facing disposition records. The r0.6 micro-repair used only the exact scientific-owner directive dated 2026-09-01 and retained the same implementation firewall. Neither revision inspected implementation. Links were provenance only and were not opened. No implementation, configuration, schema, test, audit, validation, reduction, generated product, history, neighboring package, external source, or model-memory source supplied scientific content. The shared core is identical to the one imported by the Scientific Rationale and Contract.